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(54) **SEMICONDUCTOR DEVICE AND FABRICATING METHOD THEREOF**

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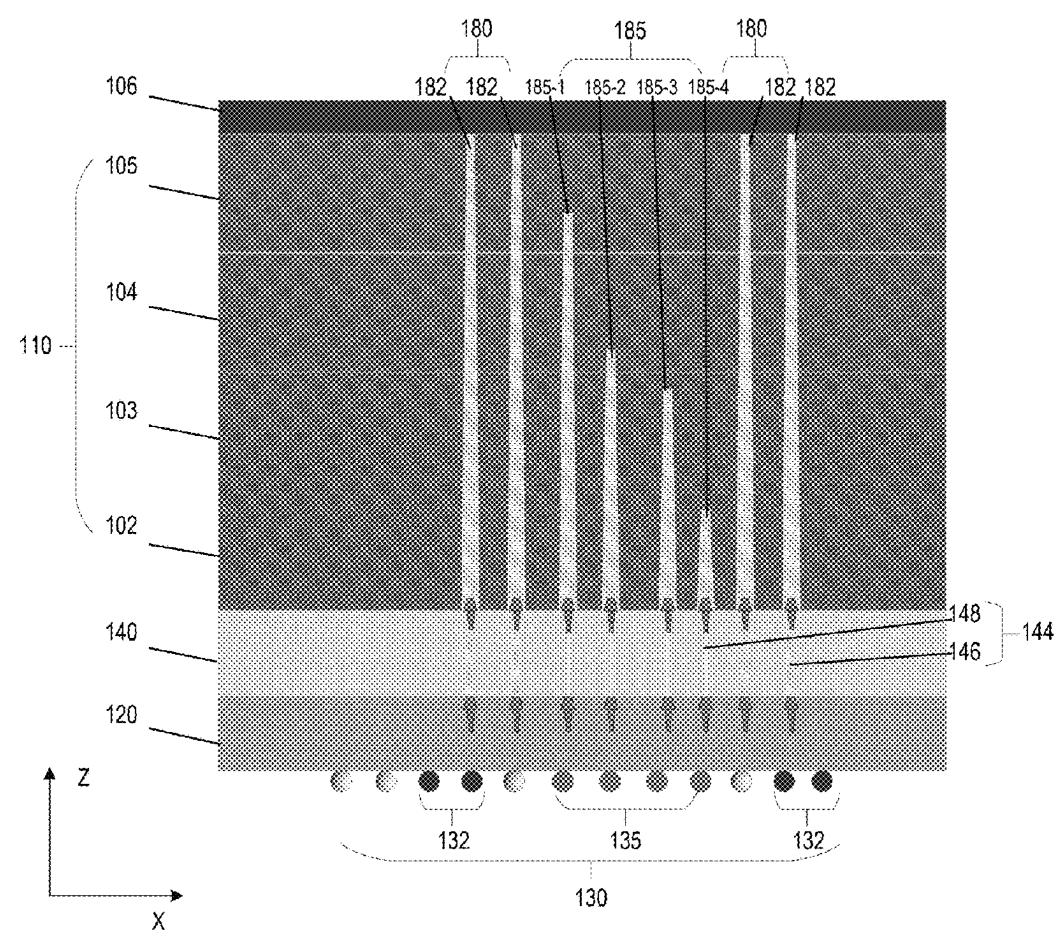
(57) **ABSTRACT**

A semiconductor device and fabricating methods thereof are disclosed. The semiconductor device includes: a plurality of dies stacked in a vertical direction; a thermal conductive layer deposited on a top surface of a topmost die; a base die located on a bottom surface of a bottommost die; and a first contact structure extending vertically through the plurality of dies. The first contact structure includes one or more first channels. Each first channel has a first end being in contact with the base die and has a second end being in contact with the thermal conductive layer.

100A



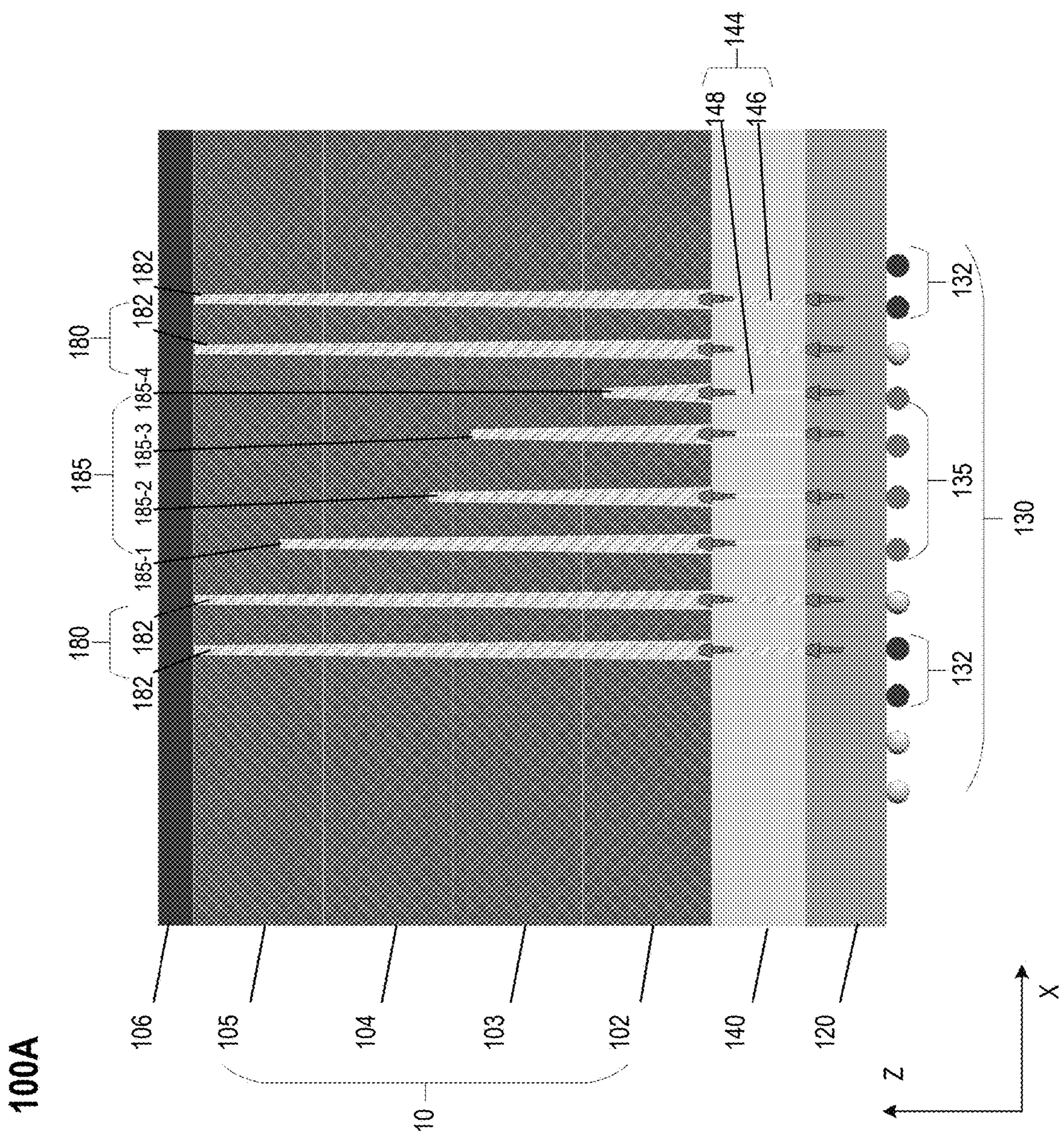

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FIG. 1A

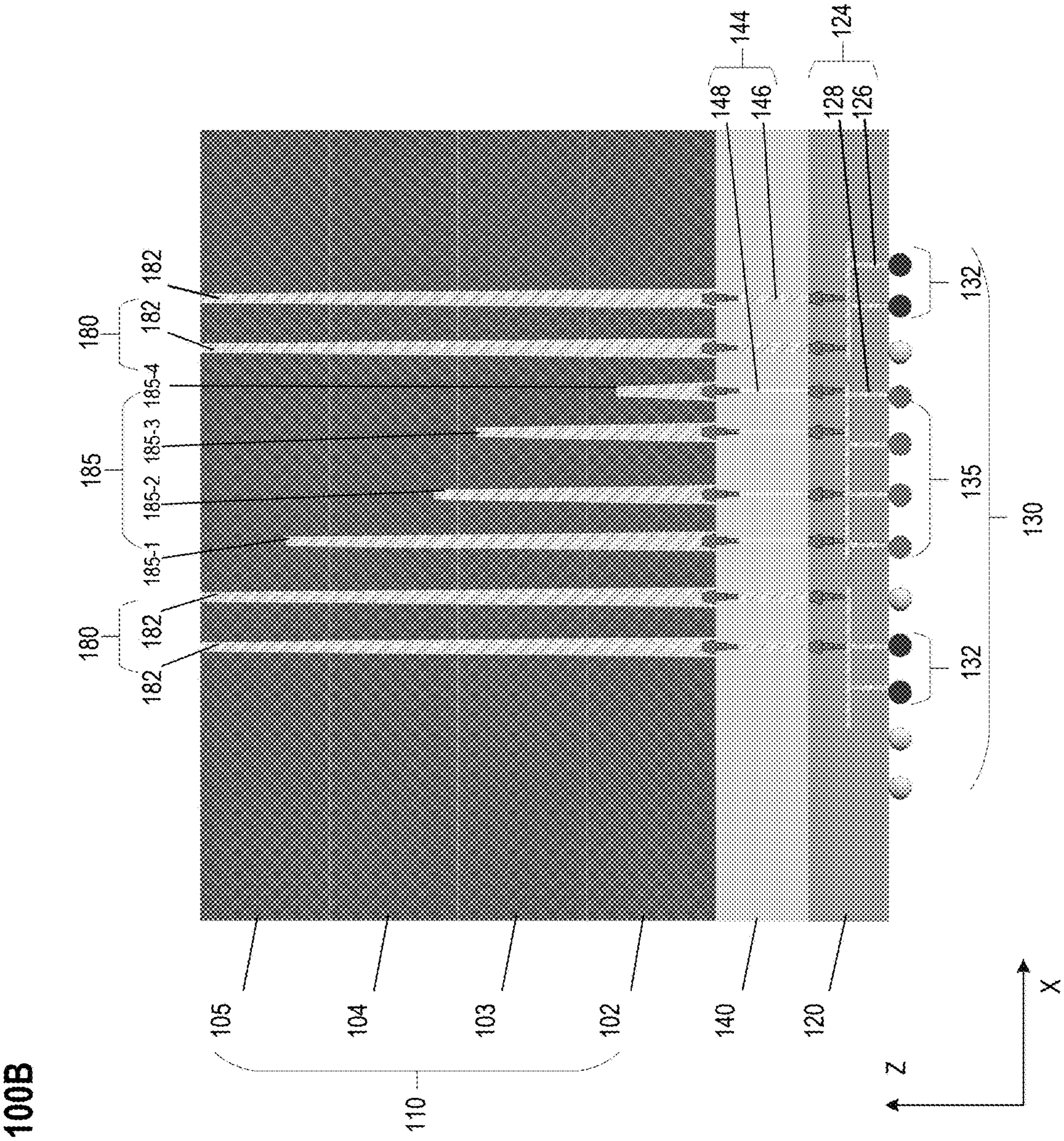
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FIG. 1B

100C

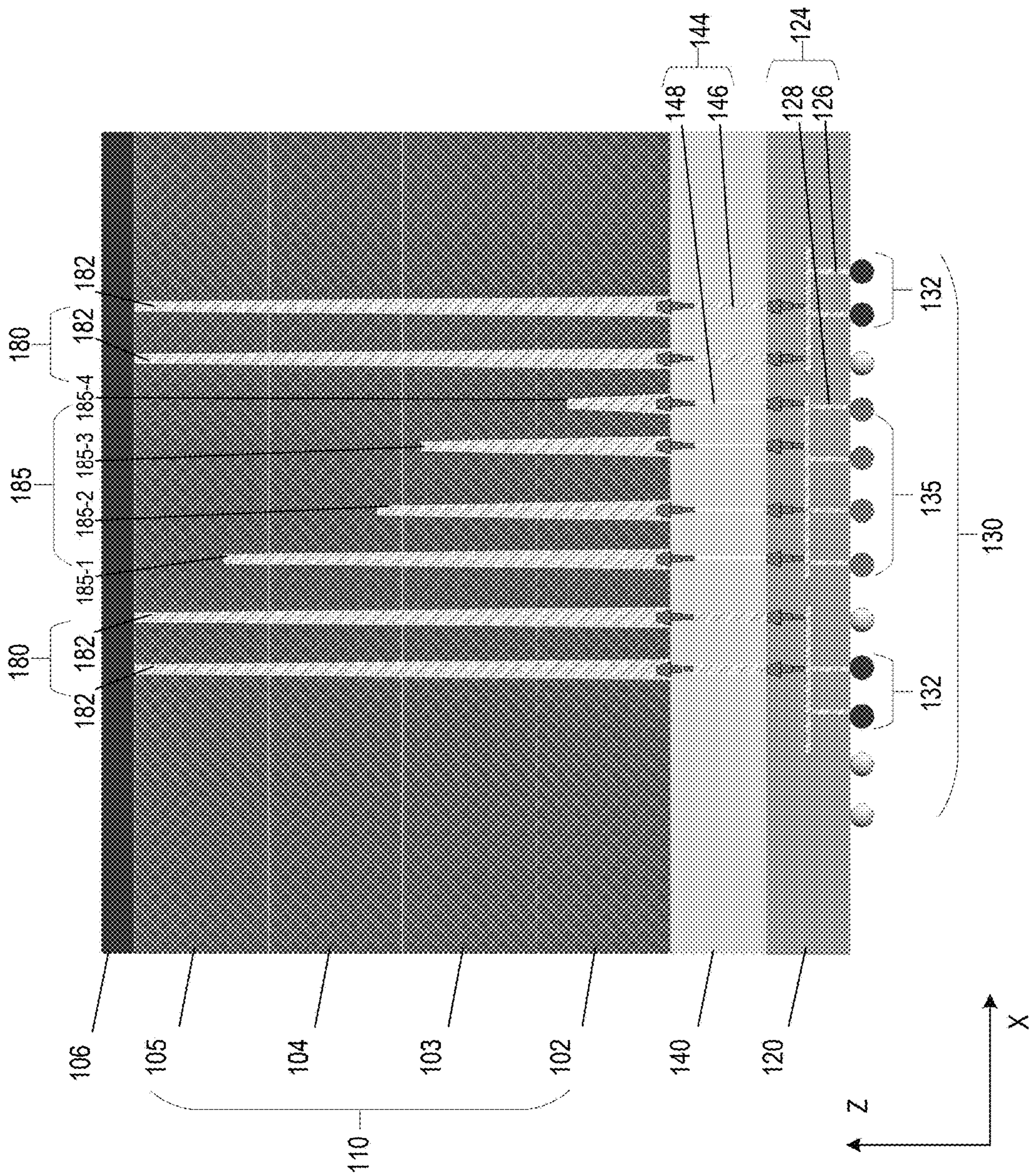

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FIG. 1C

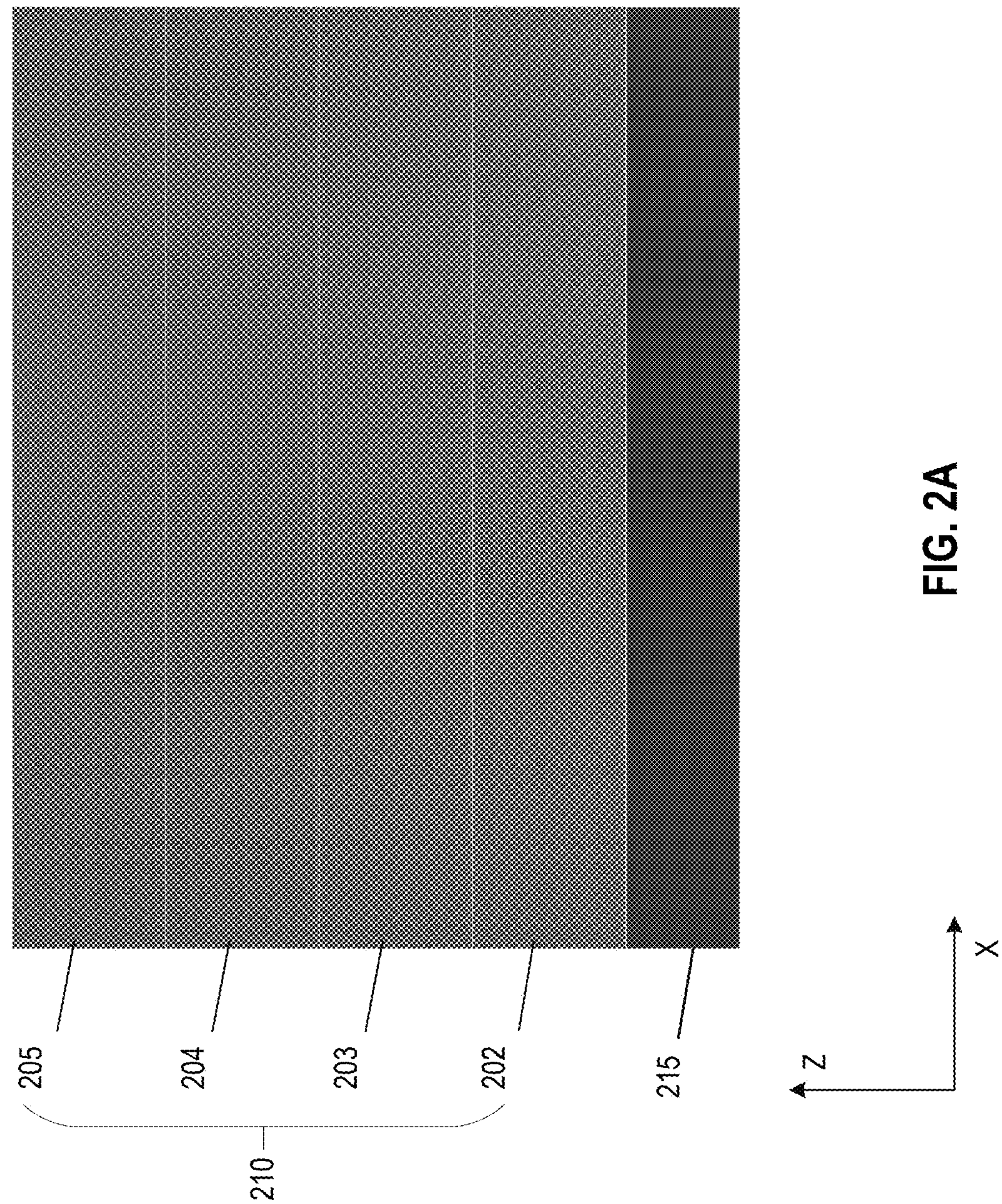

FIG. 2A

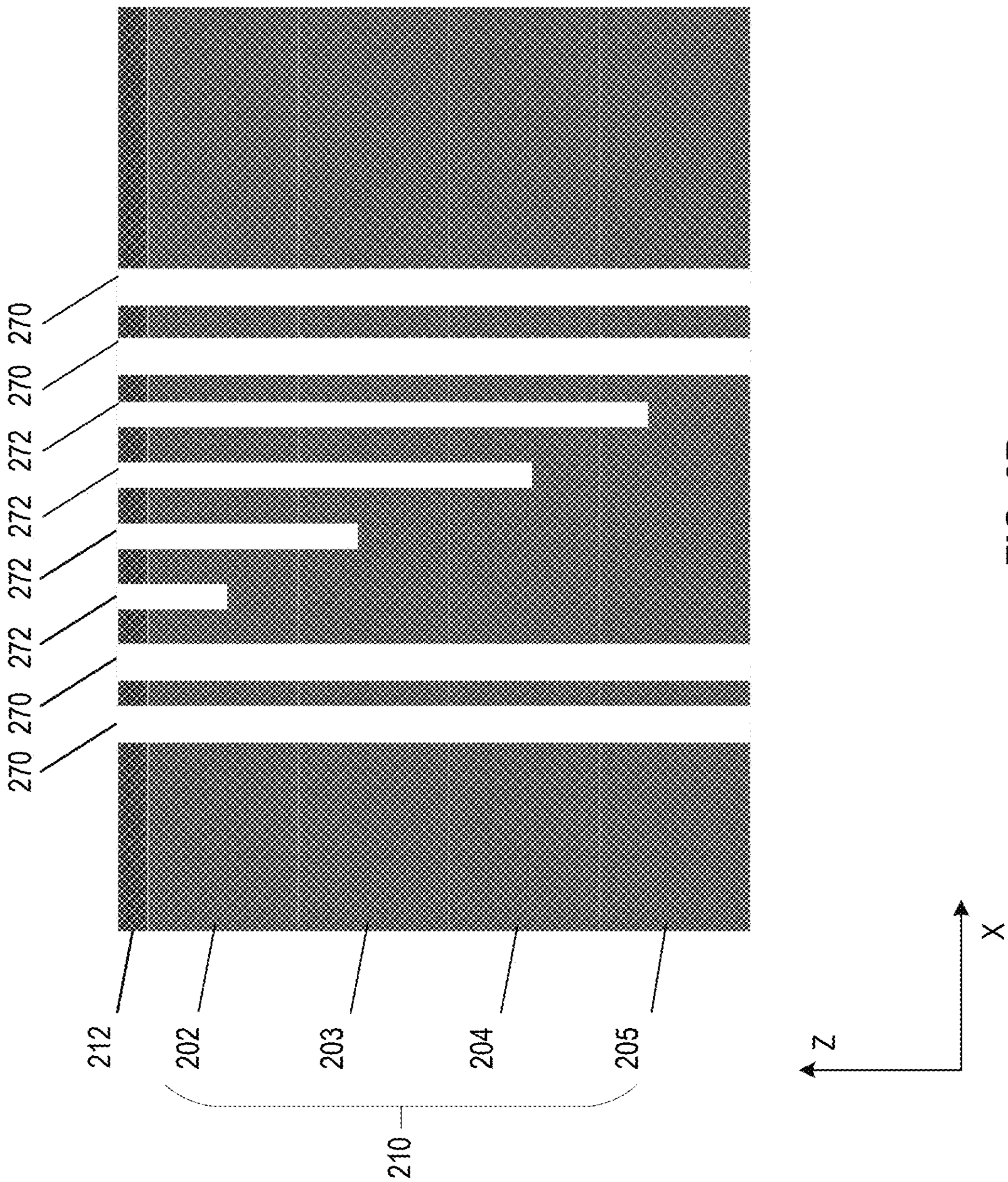

FIG. 2B

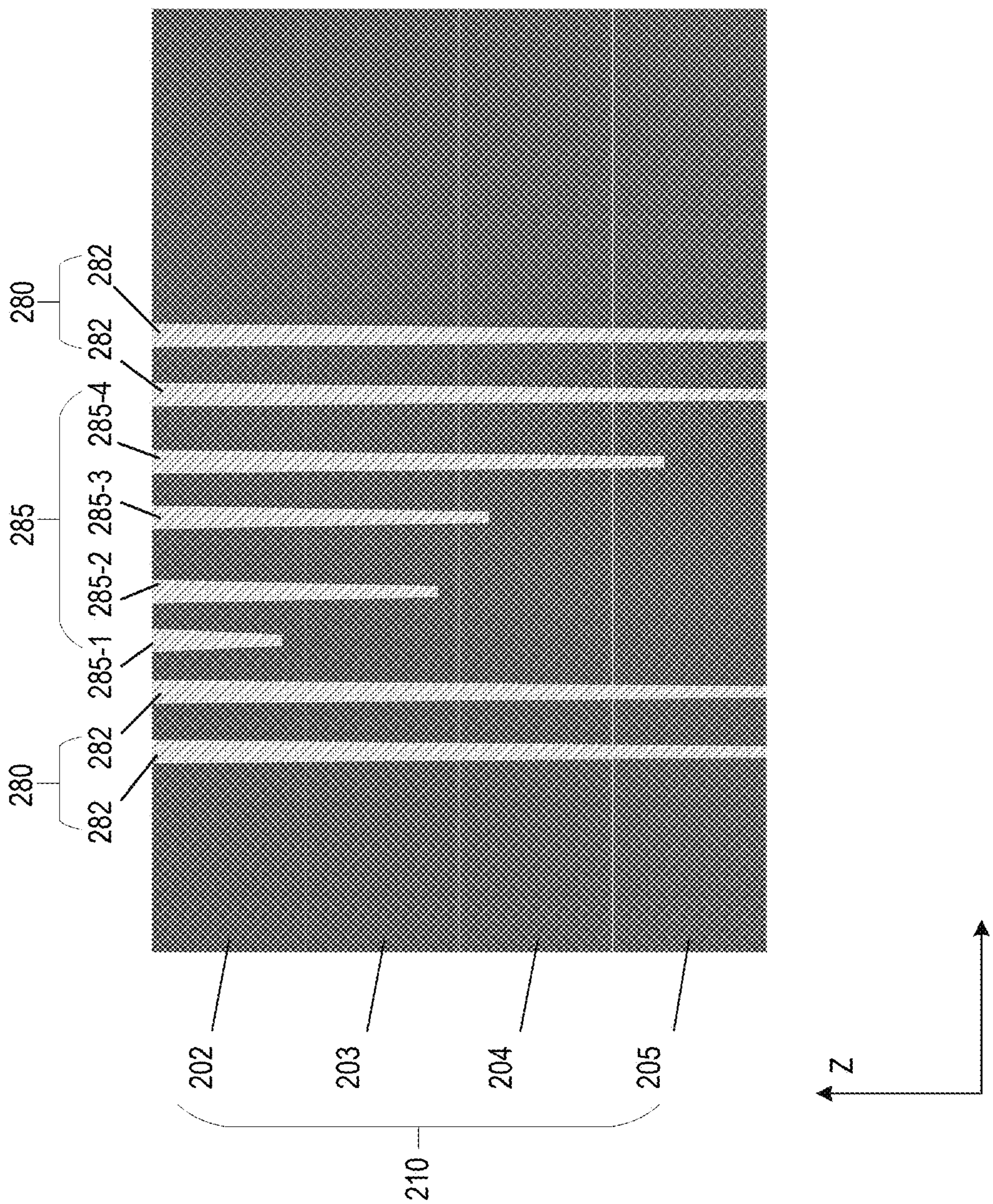
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FIG. 2C

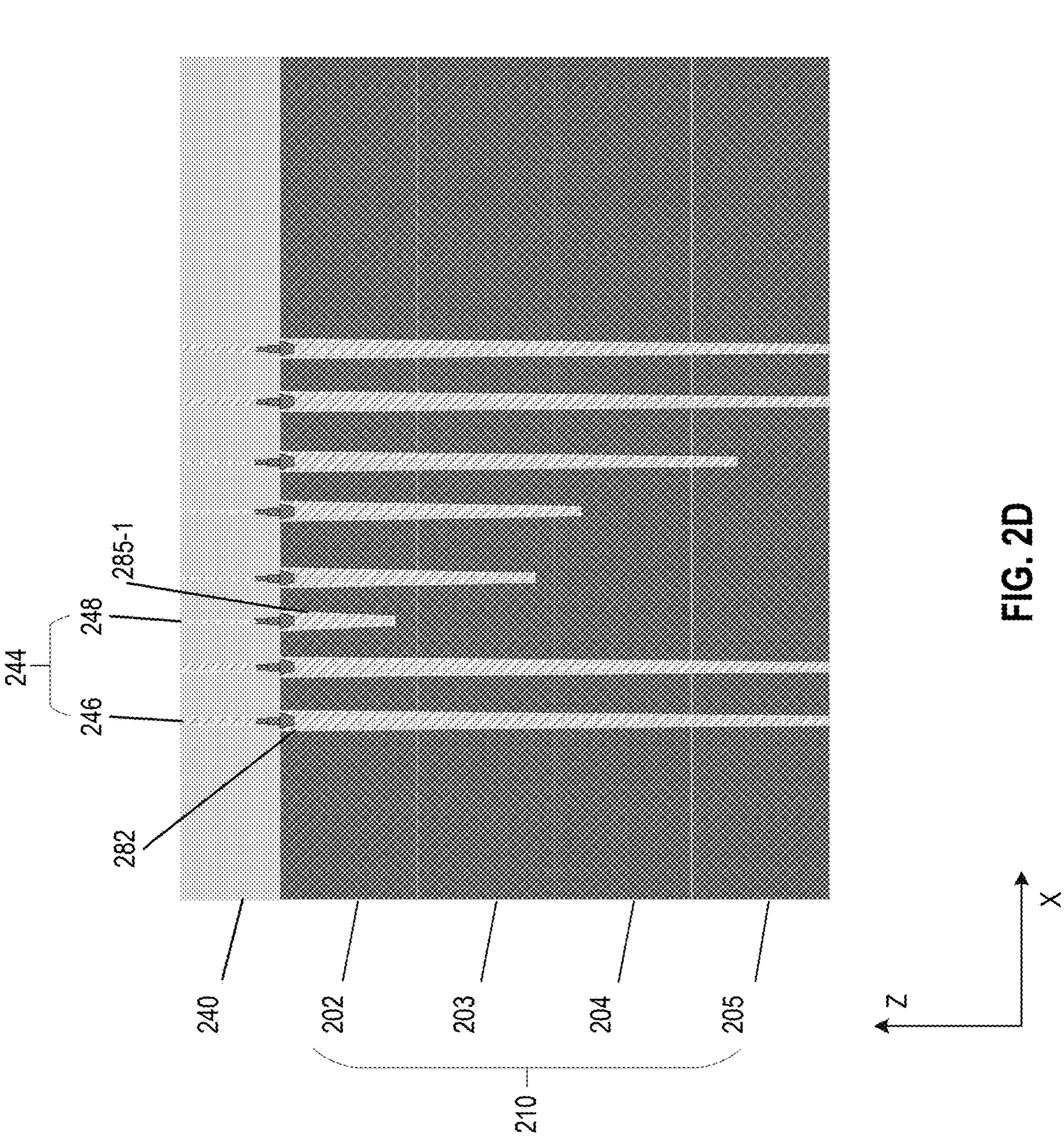

FIG. 2D

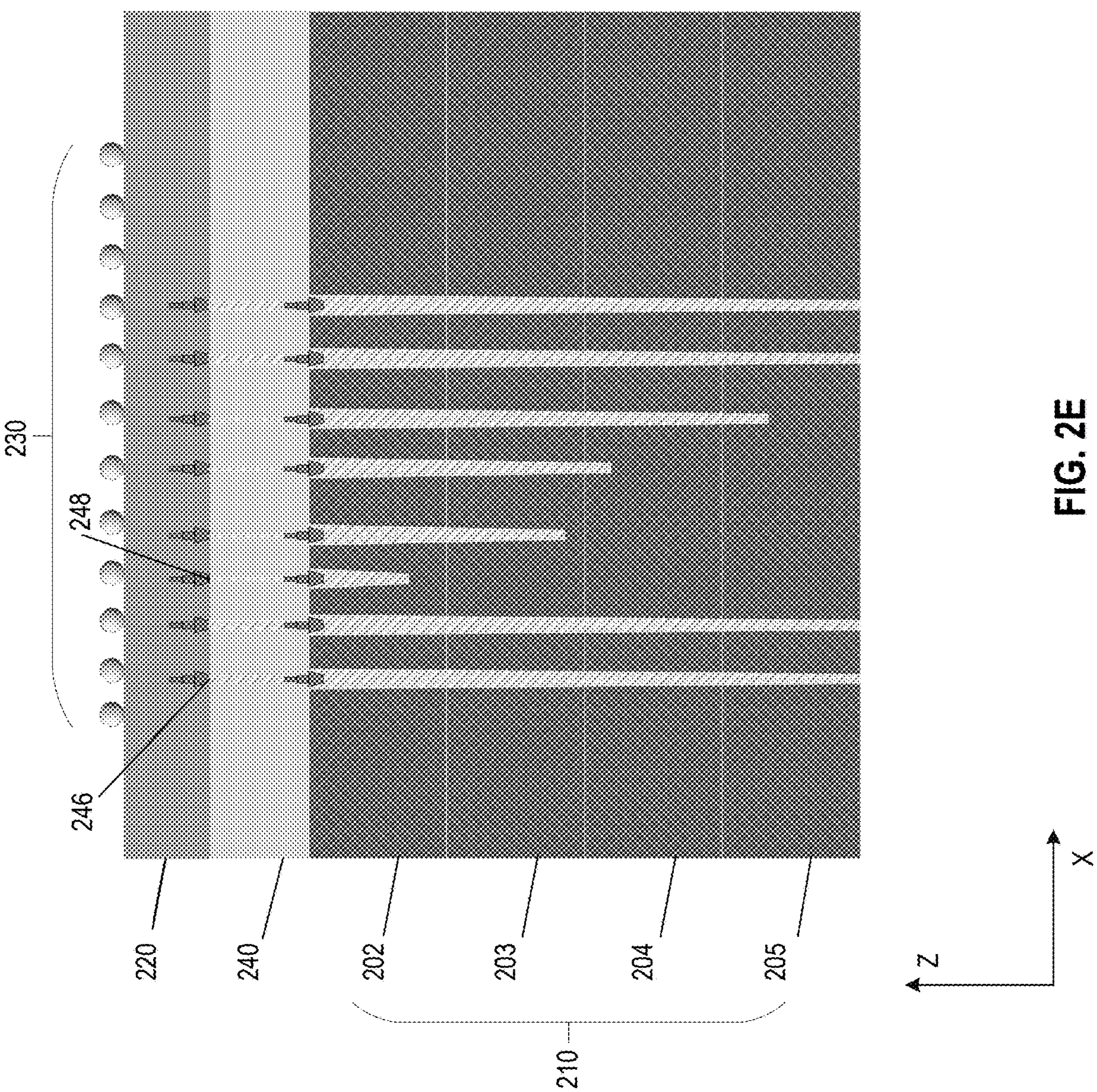

FIG. 2E

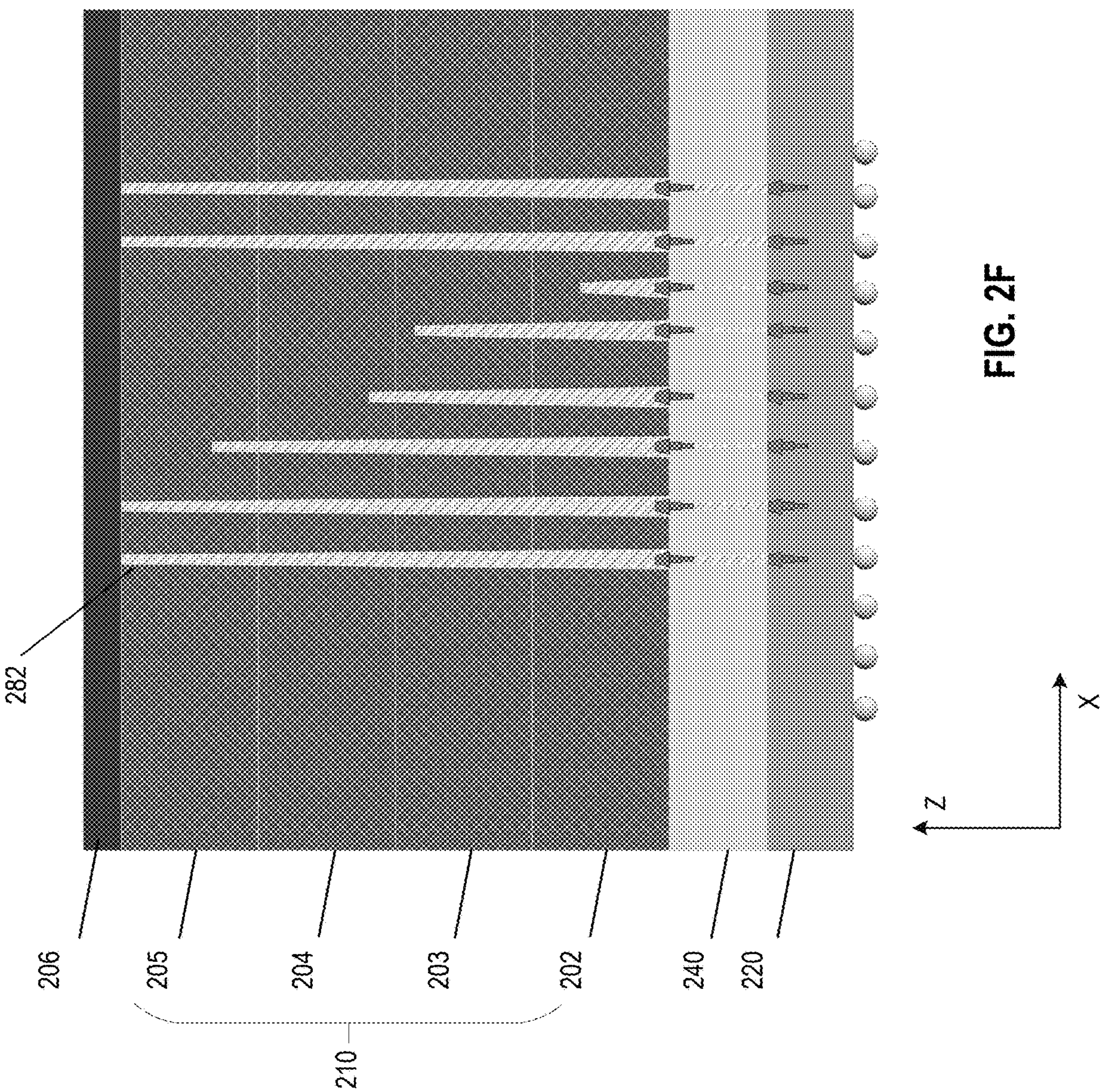

FIG. 2F

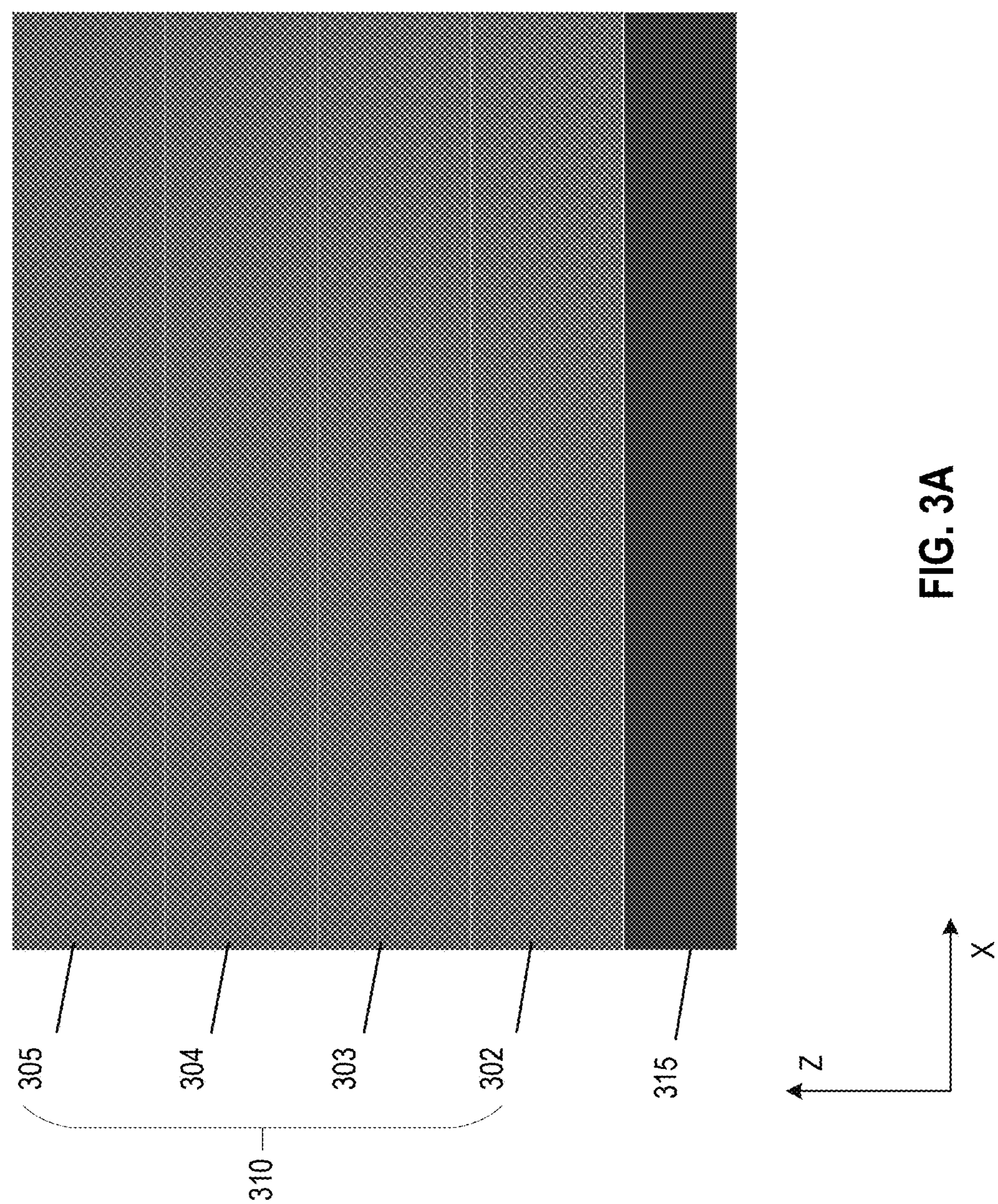

FIG. 3A

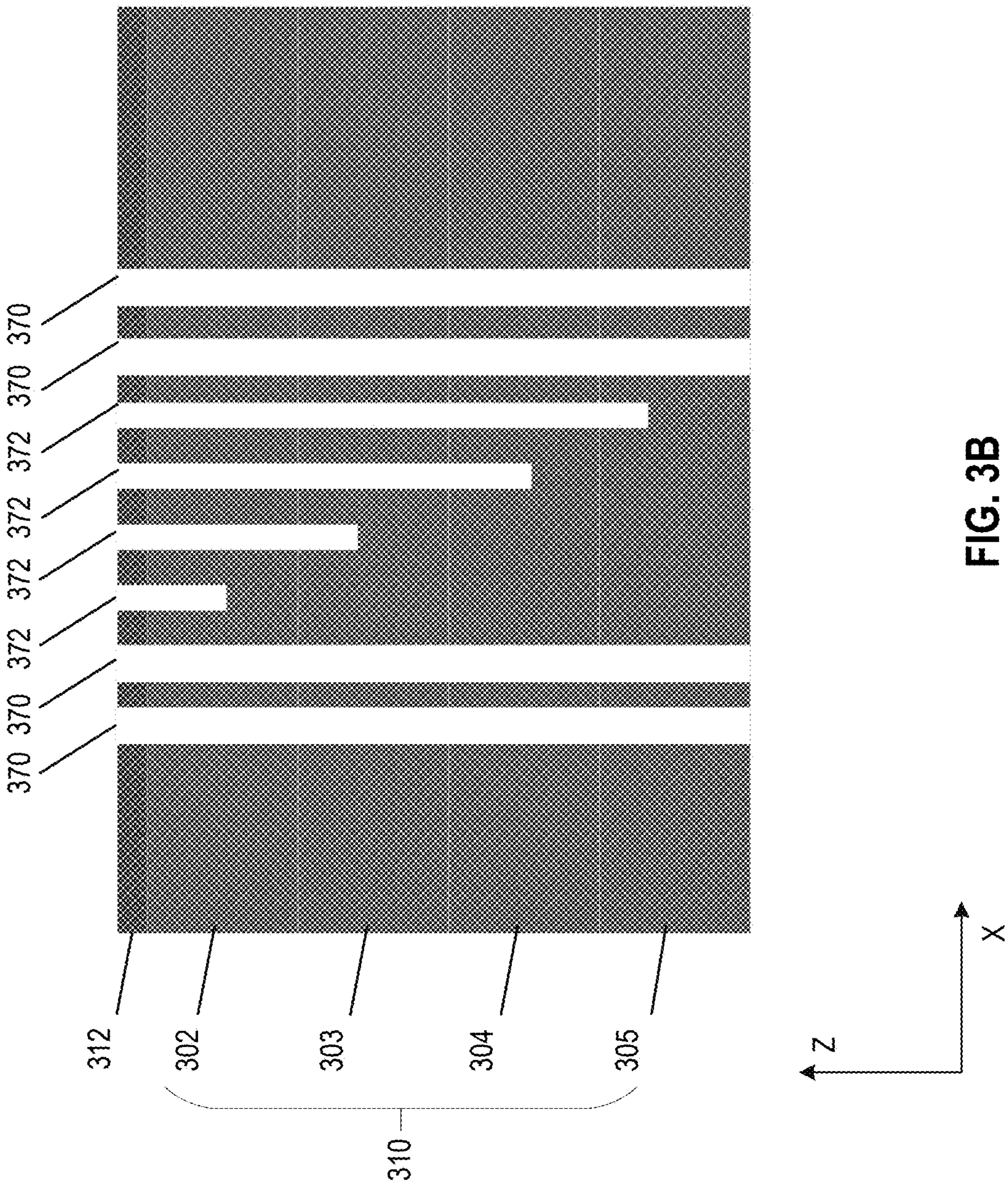

FIG. 3B

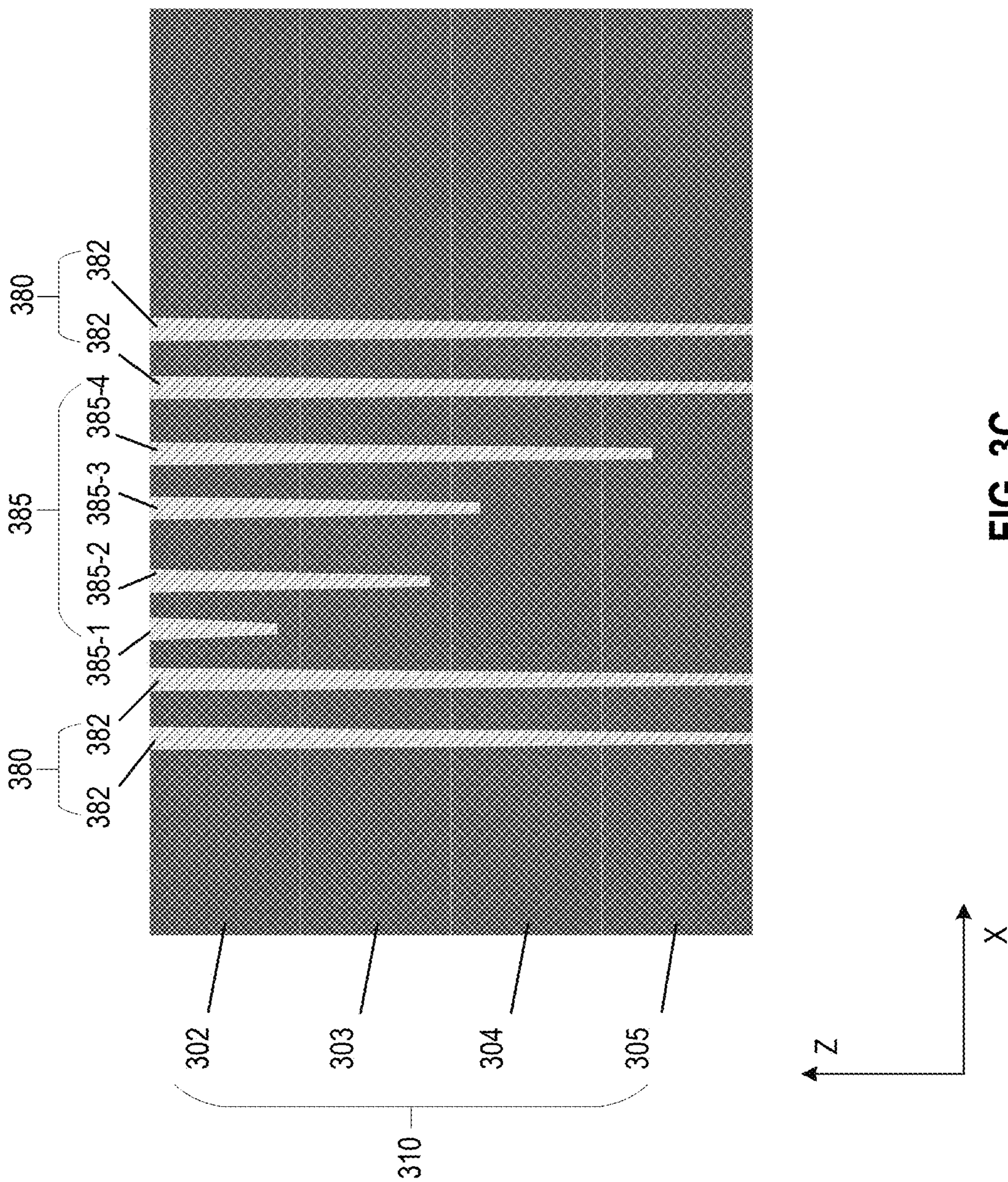
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FIG. 3C

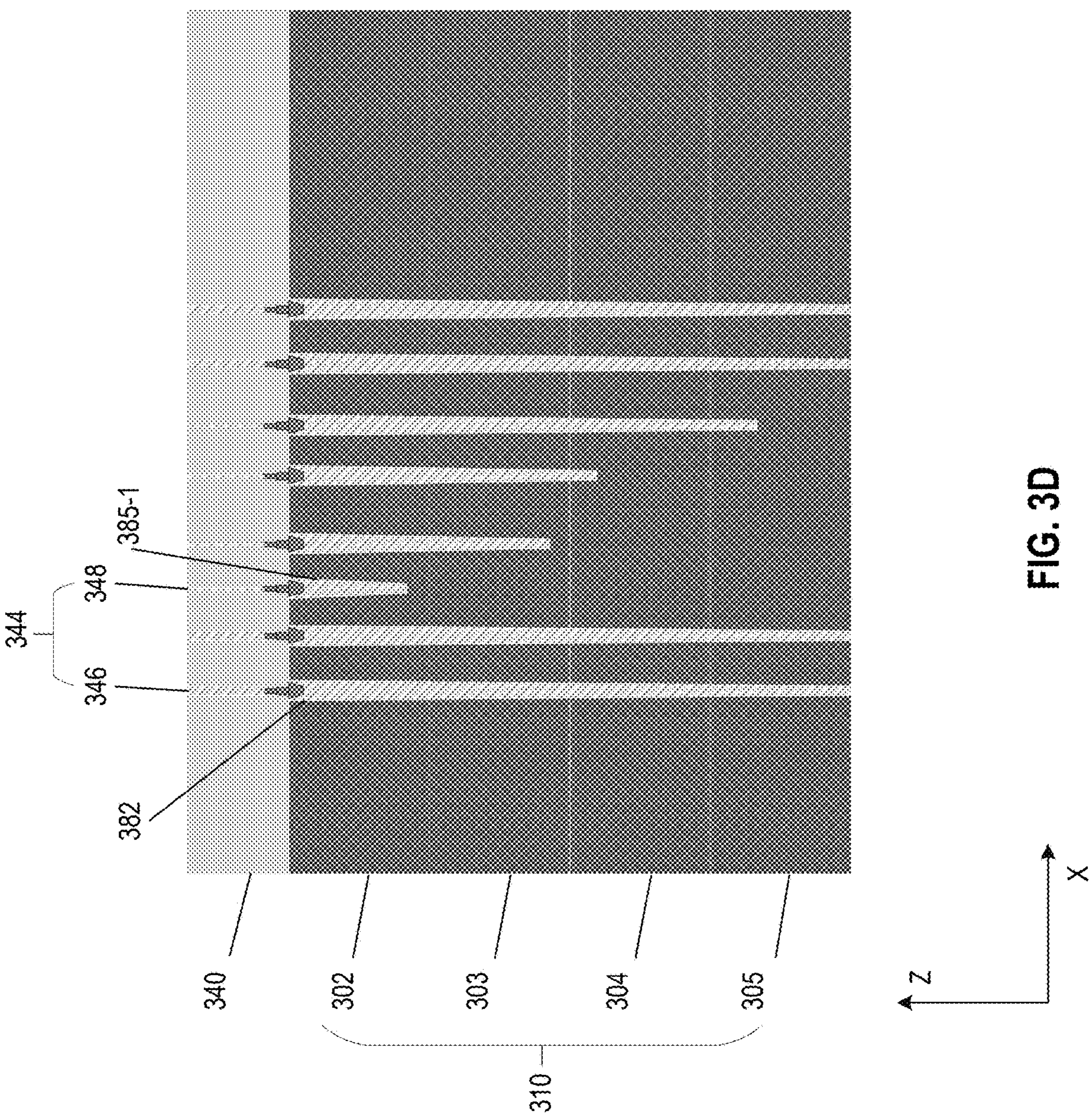

FIG. 3D

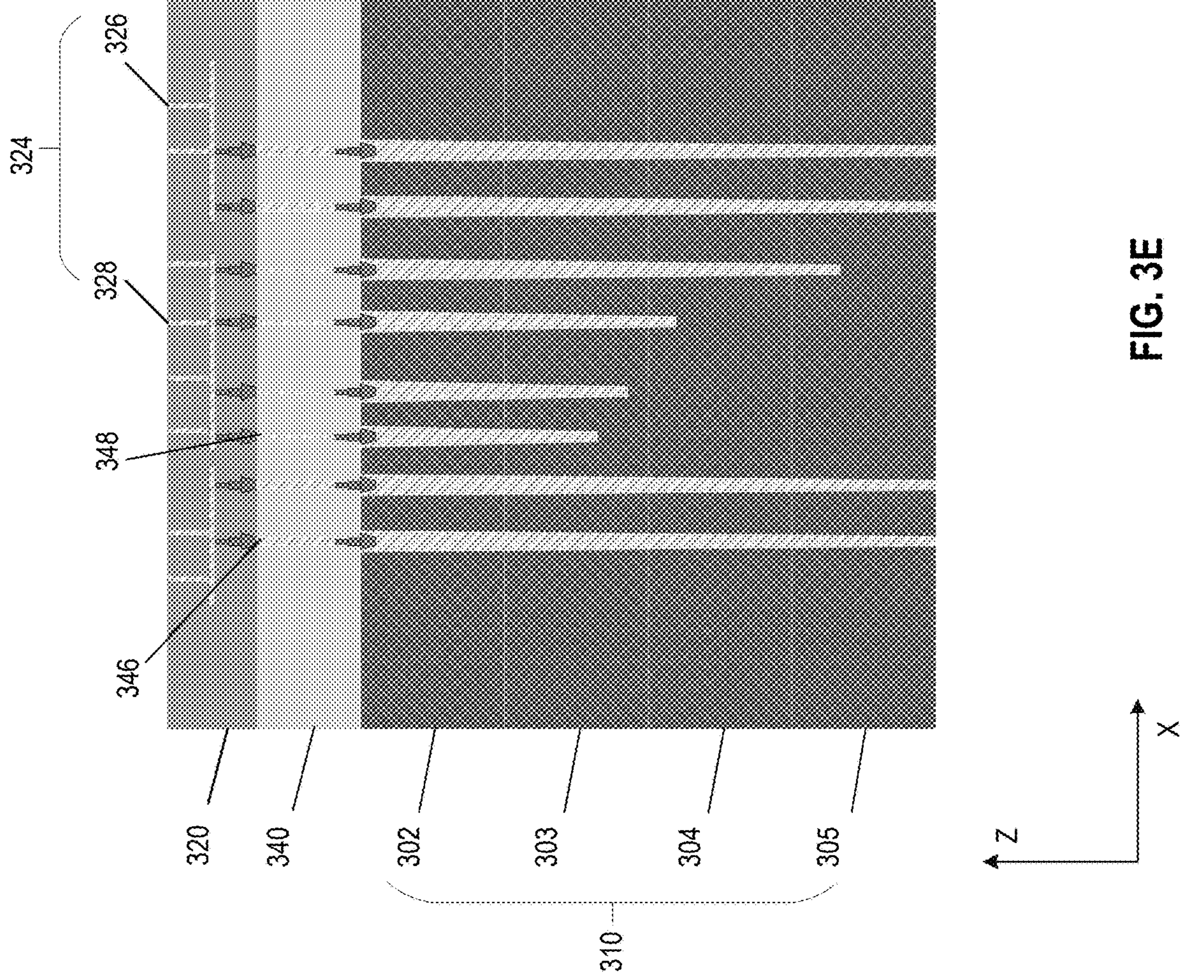

FIG. 3E

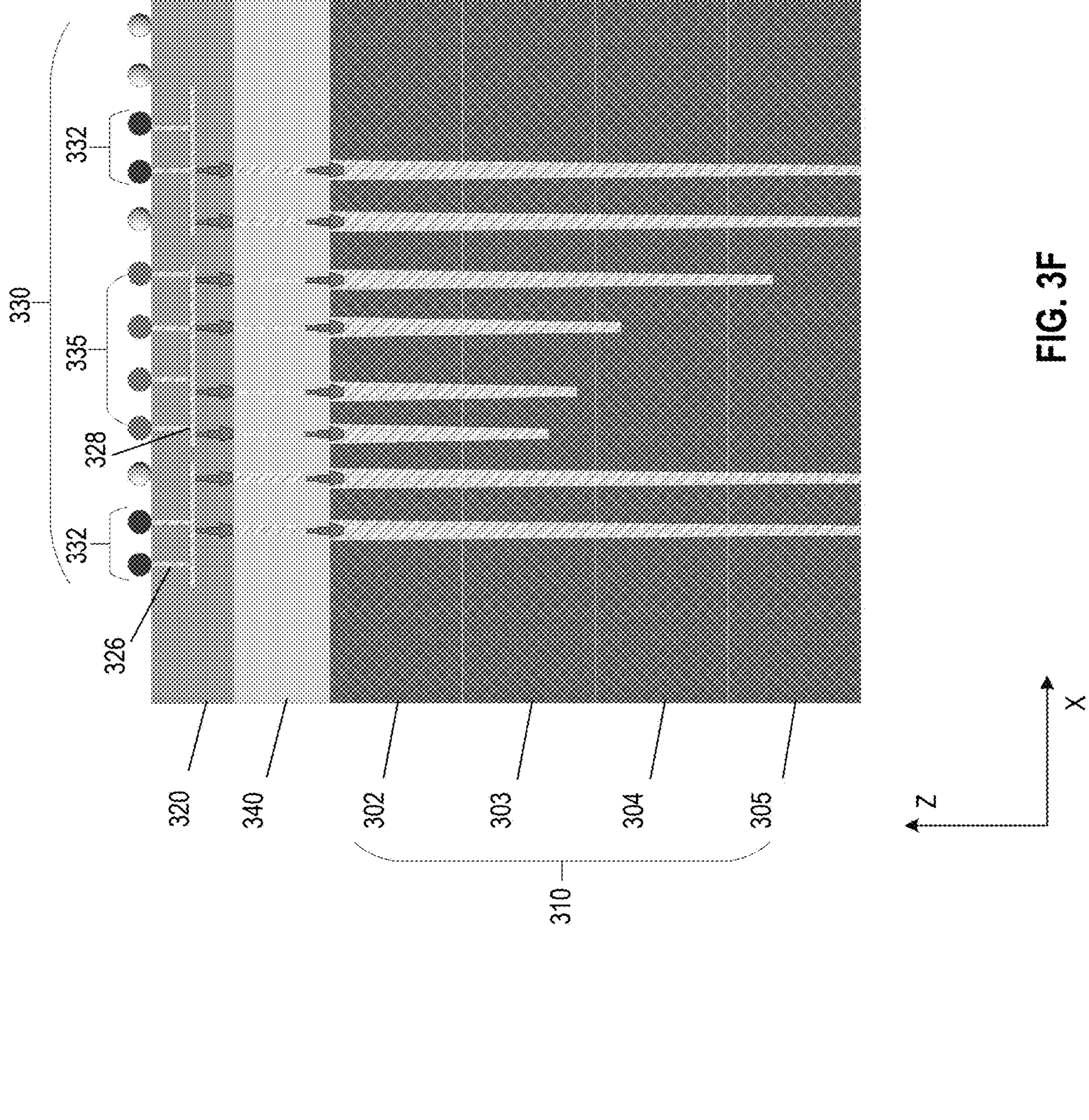

FIG. 3F

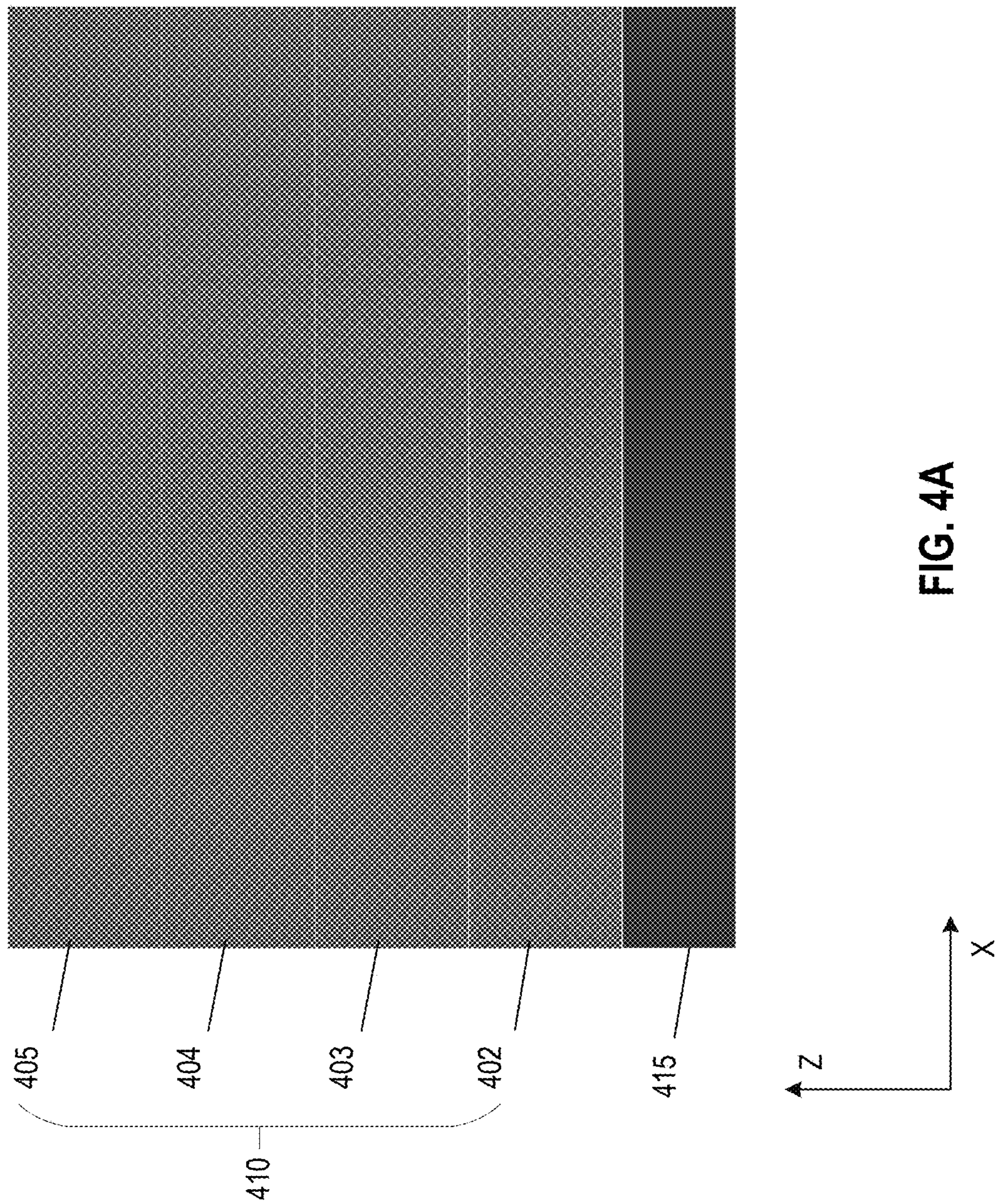

FIG. 4A

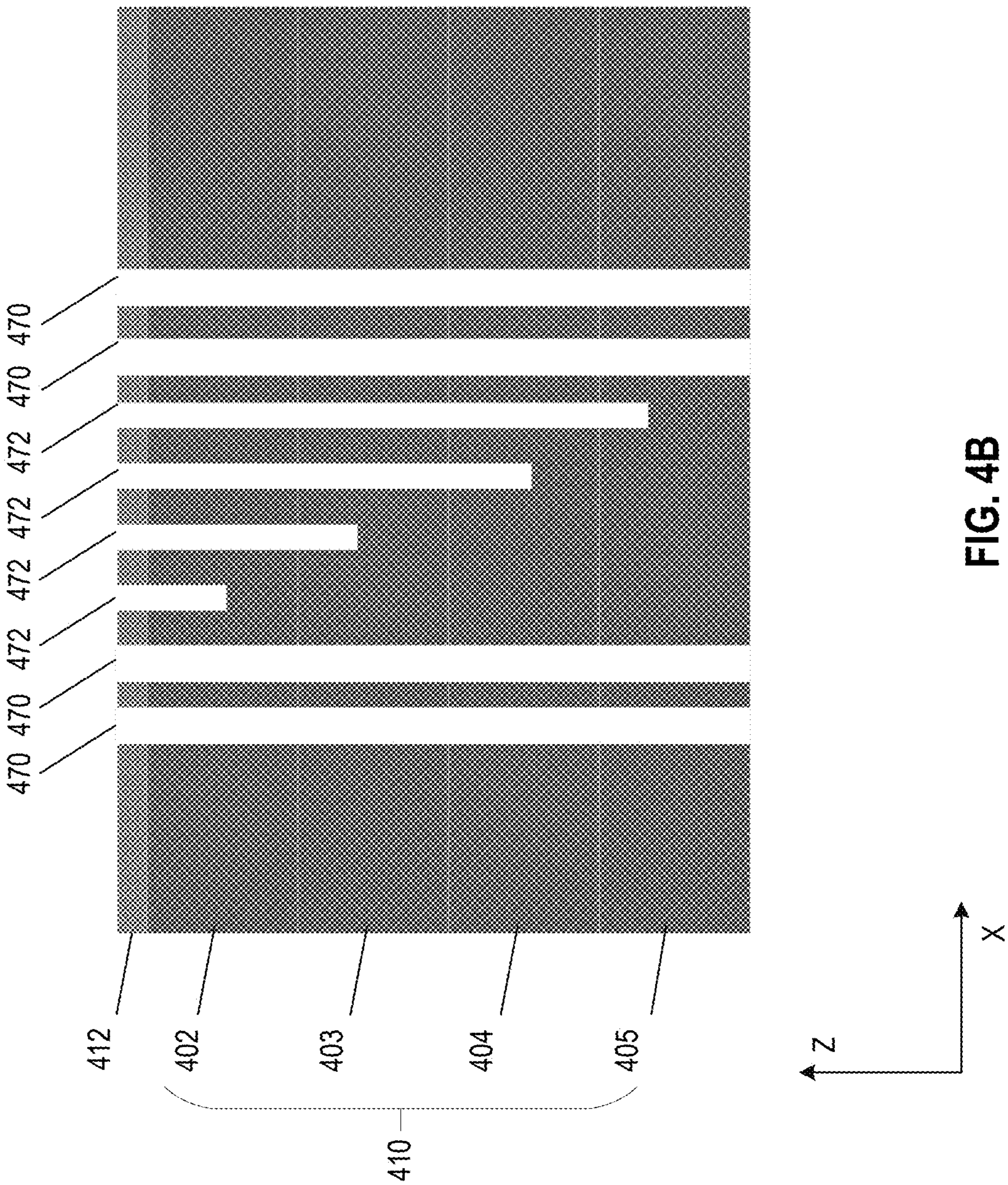

FIG. 4B

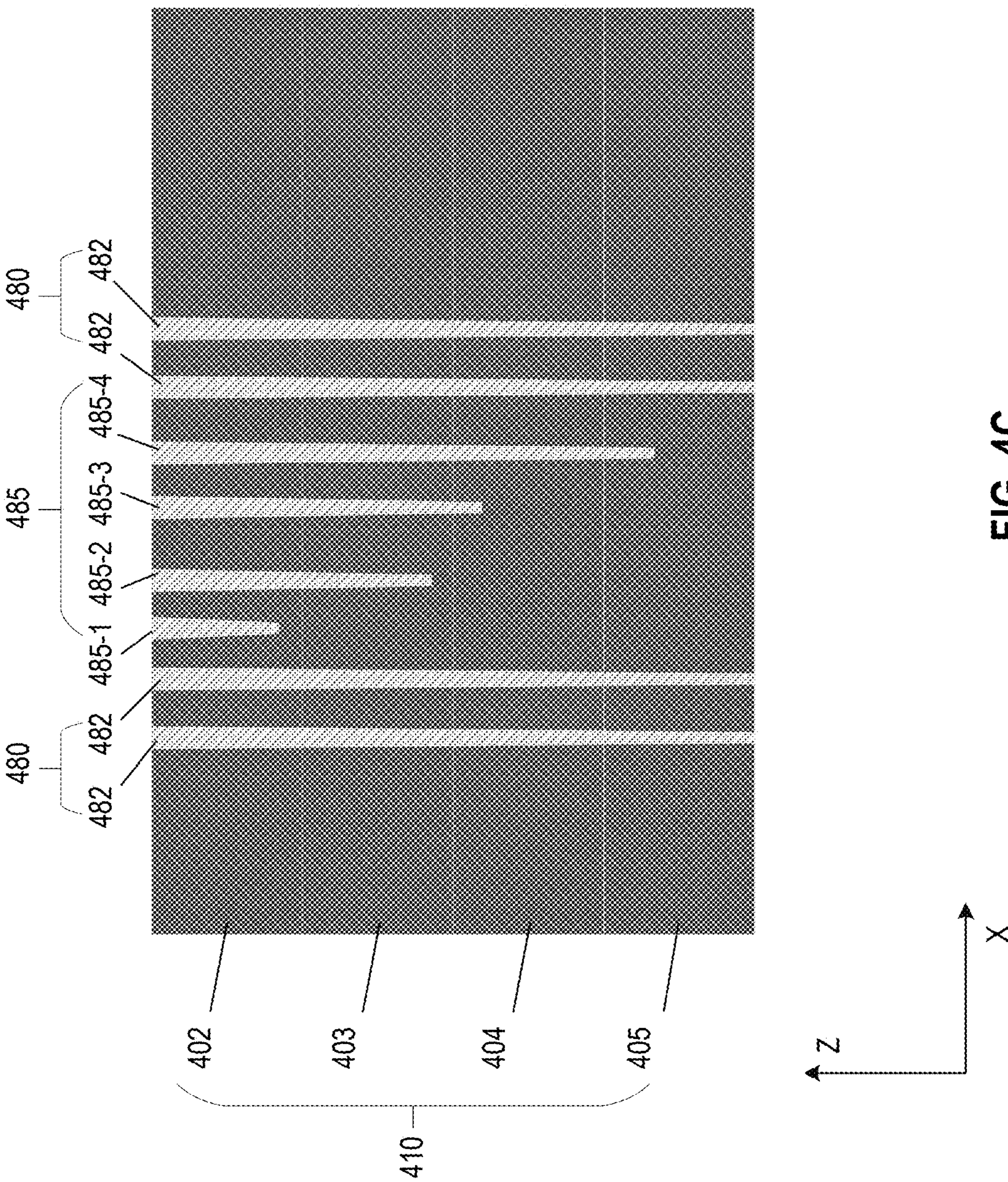
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FIG. 4C

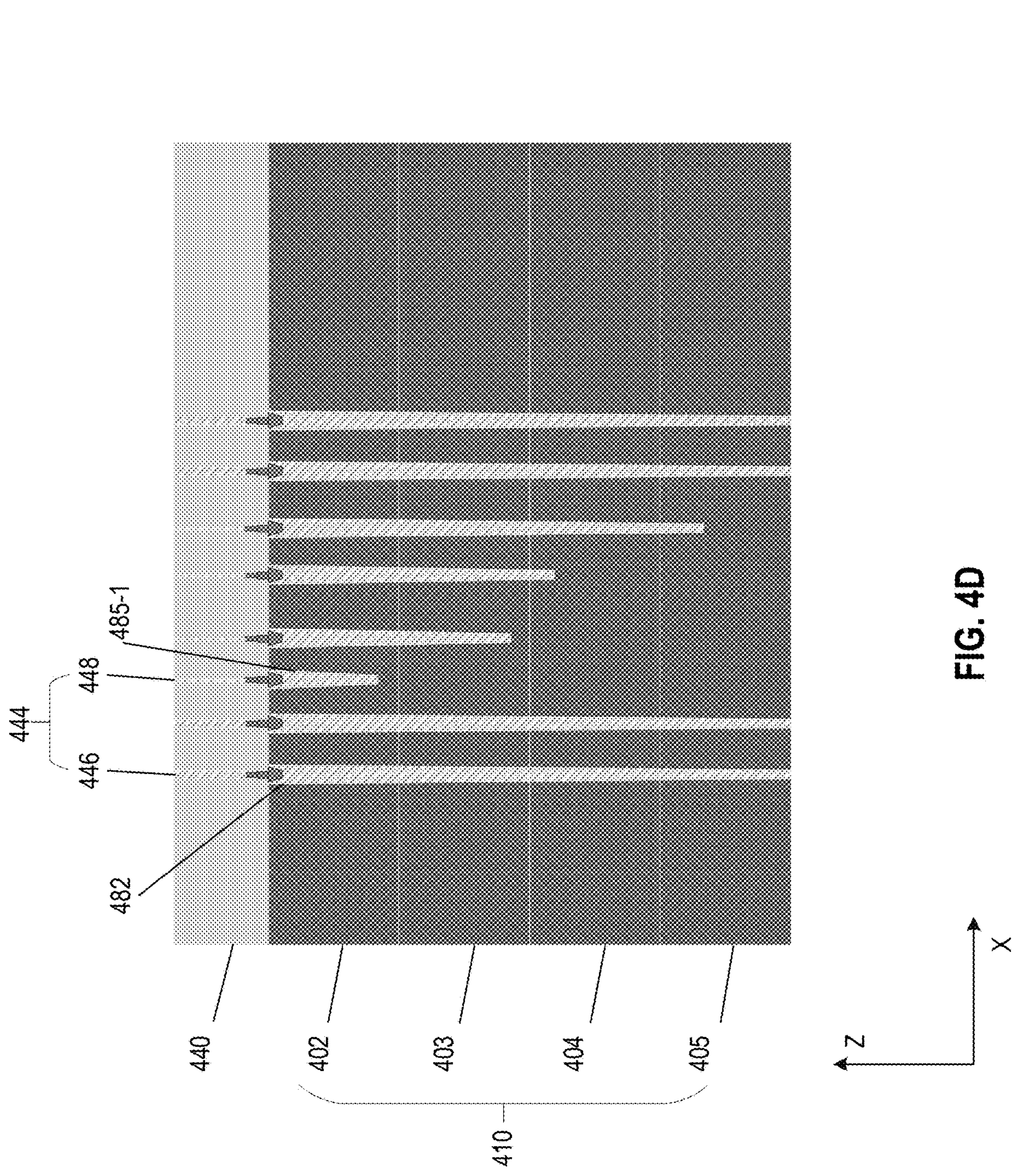

FIG. 4D

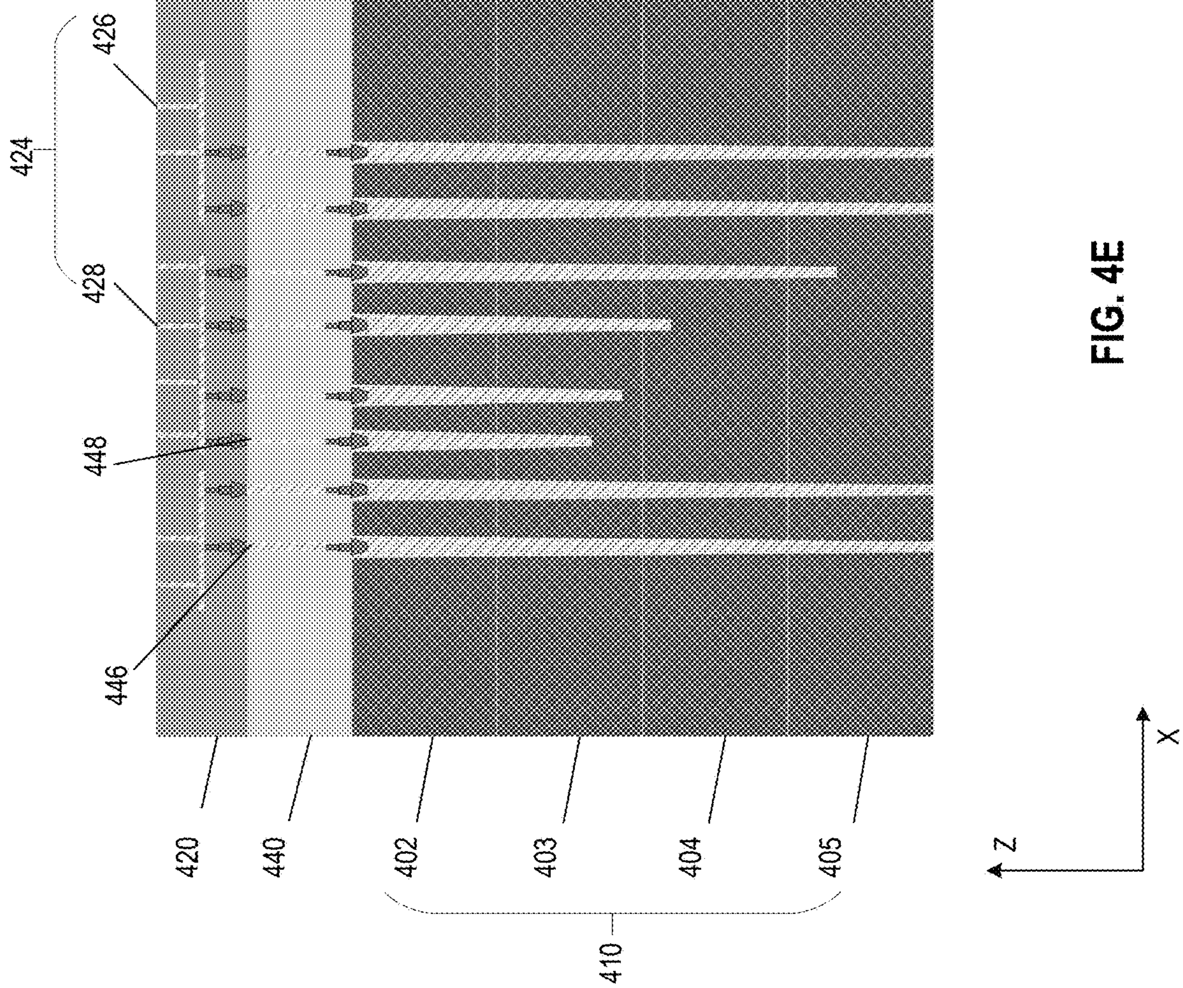

FIG. 4E

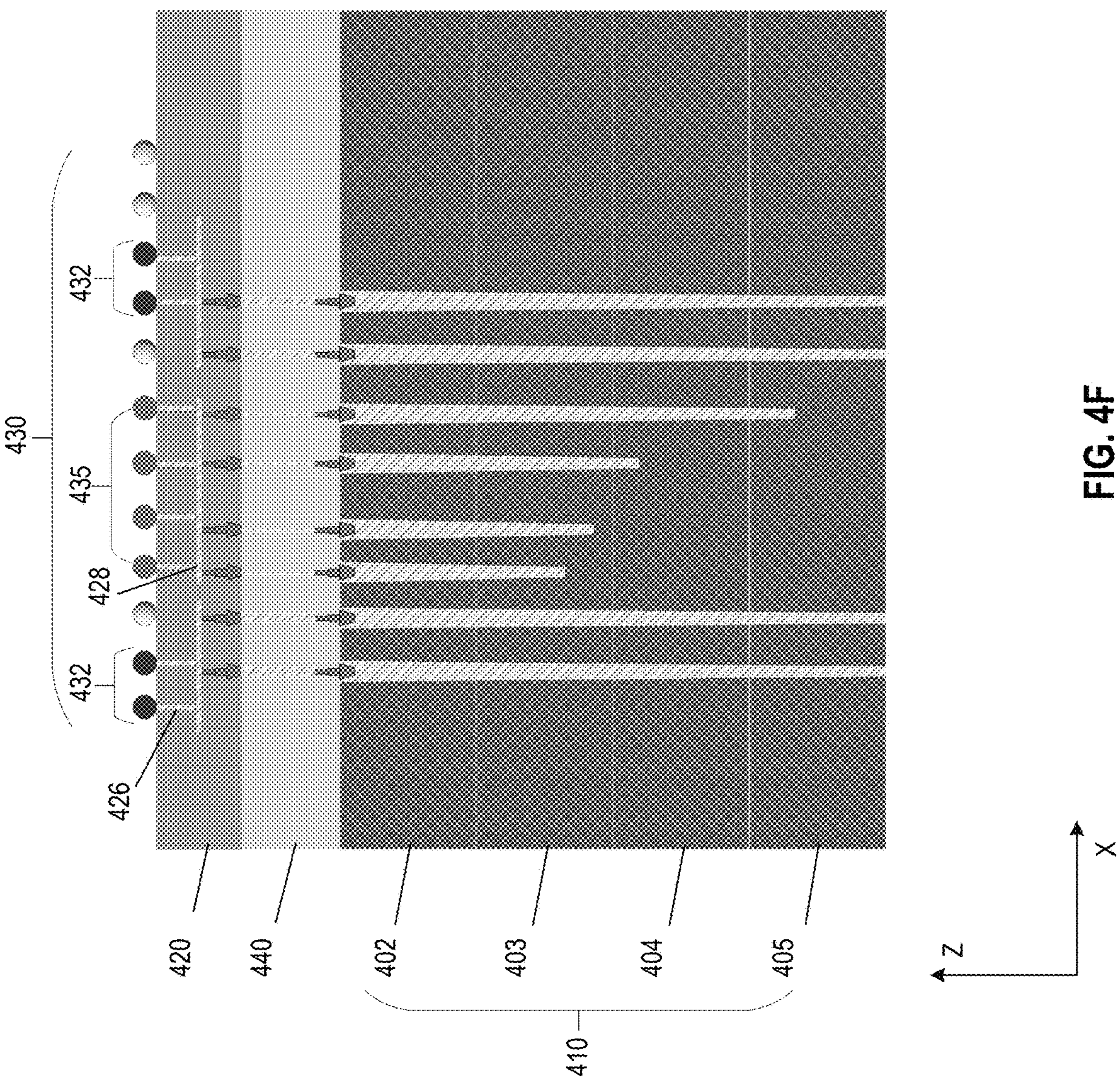

FIG. 4F

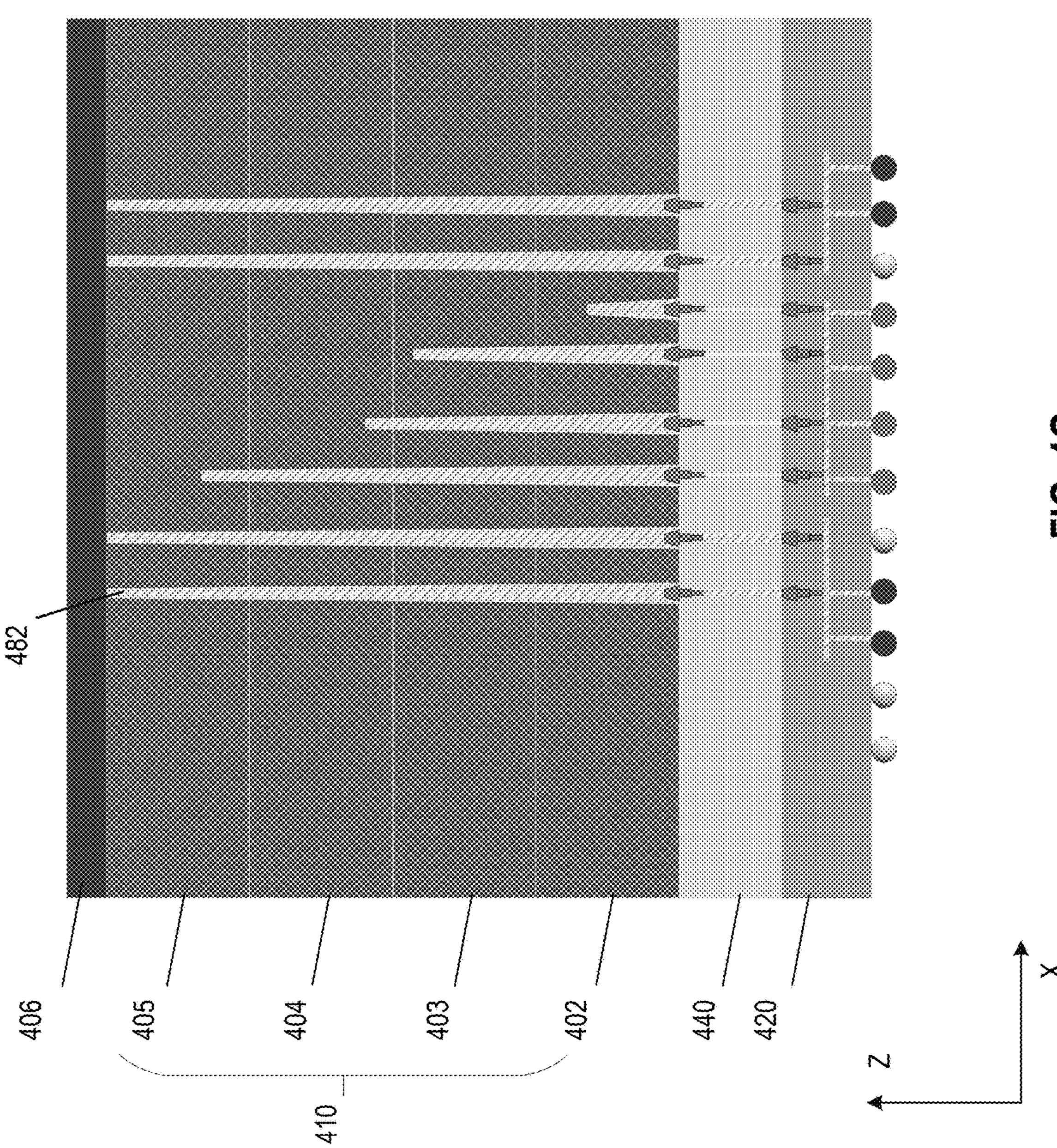

FIG. 4G

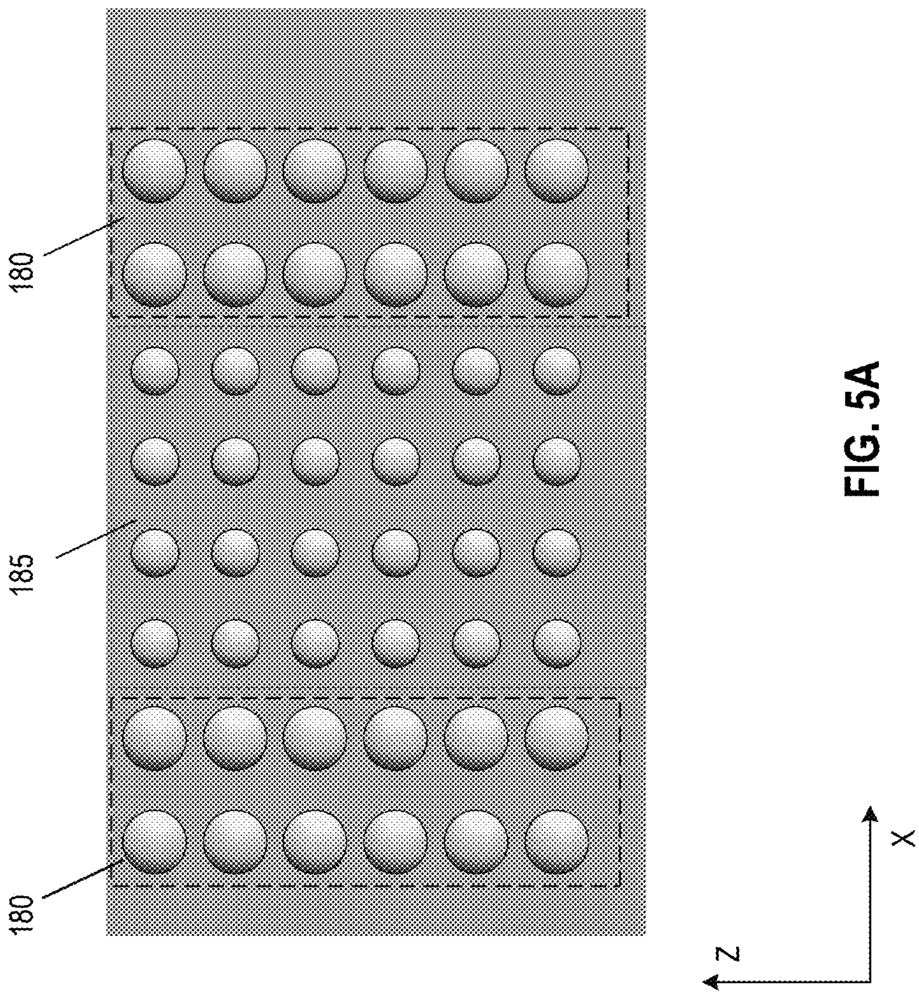

FIG. 5A

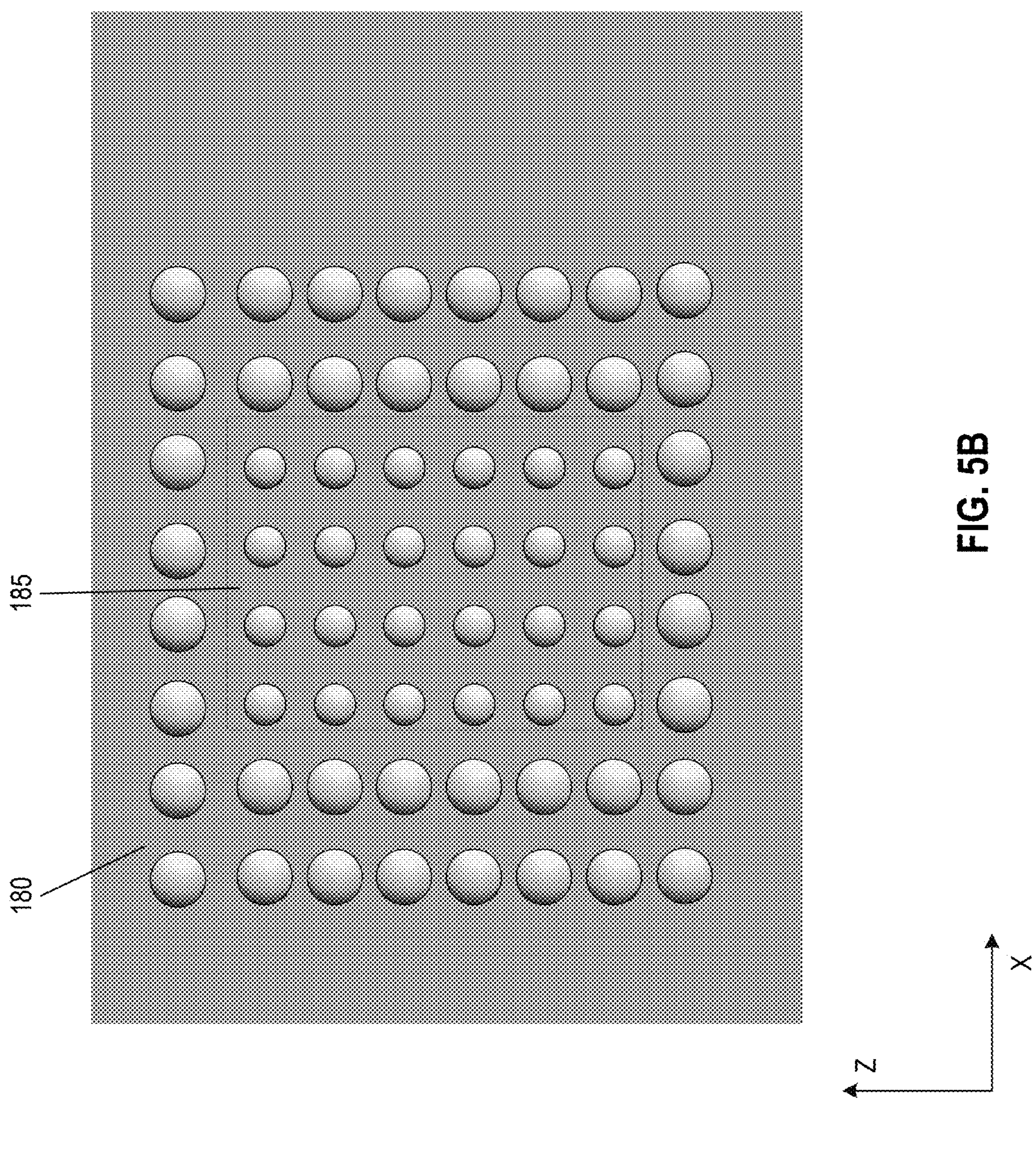

FIG. 5B

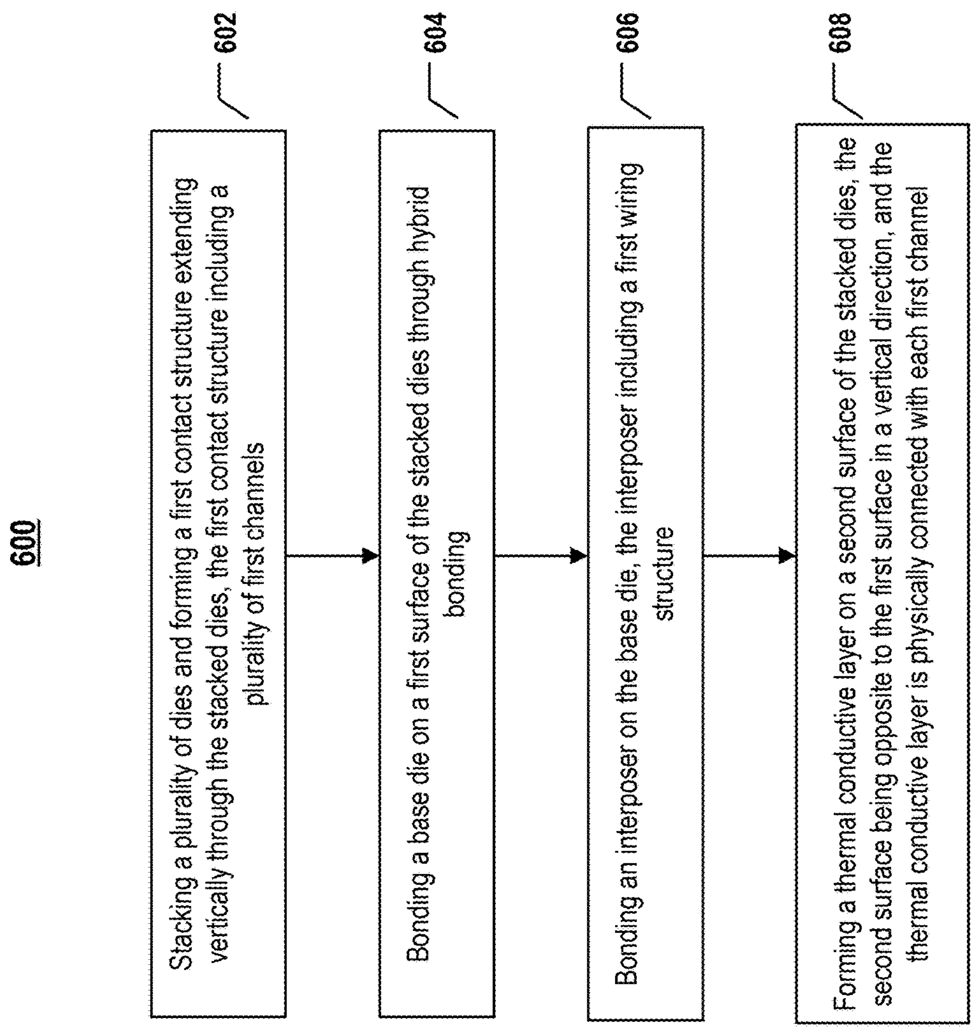
600
602
Stacking a plurality of dies and forming a first contact structure extending vertically through the stacked dies, the first contact structure including a plurality of first channels
604
Bonding a base die on a first surface of the stacked dies through hybrid bonding
606
Bonding an interposer on the base die, the interposer including a first wiring structure
608
Forming a thermal conductive layer on a second surface of the stacked dies, the second surface being opposite to the first surface in a vertical direction, and the thermal conductive layer is physically connected with each first channel

FIG. 6

700

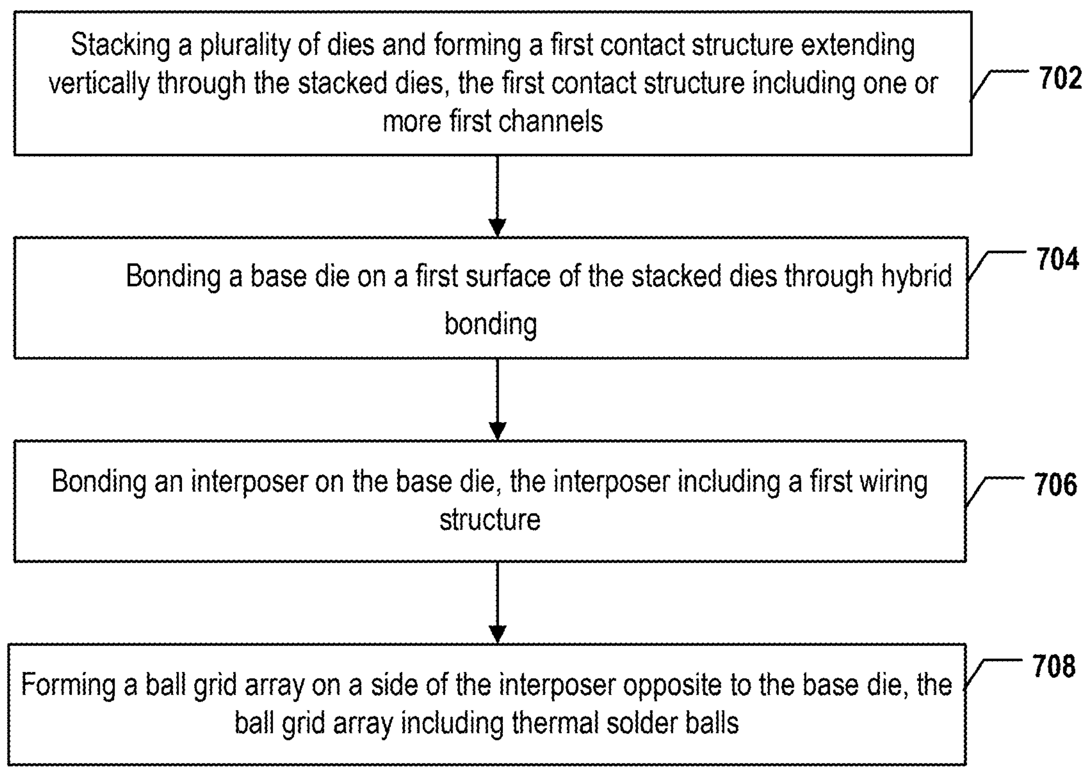
Stacking a plurality of dies and forming a first contact structure extending vertically through the stacked dies, the first contact structure including one or more first channels
702
Bonding a base die on a first surface of the stacked dies through hybrid bonding
704
Bonding an interposer on the base die, the interposer including a first wiring structure
706
Forming a ball grid array on a side of the interposer opposite to the base die, the ball grid array including thermal solder balls
708

FIG. 7

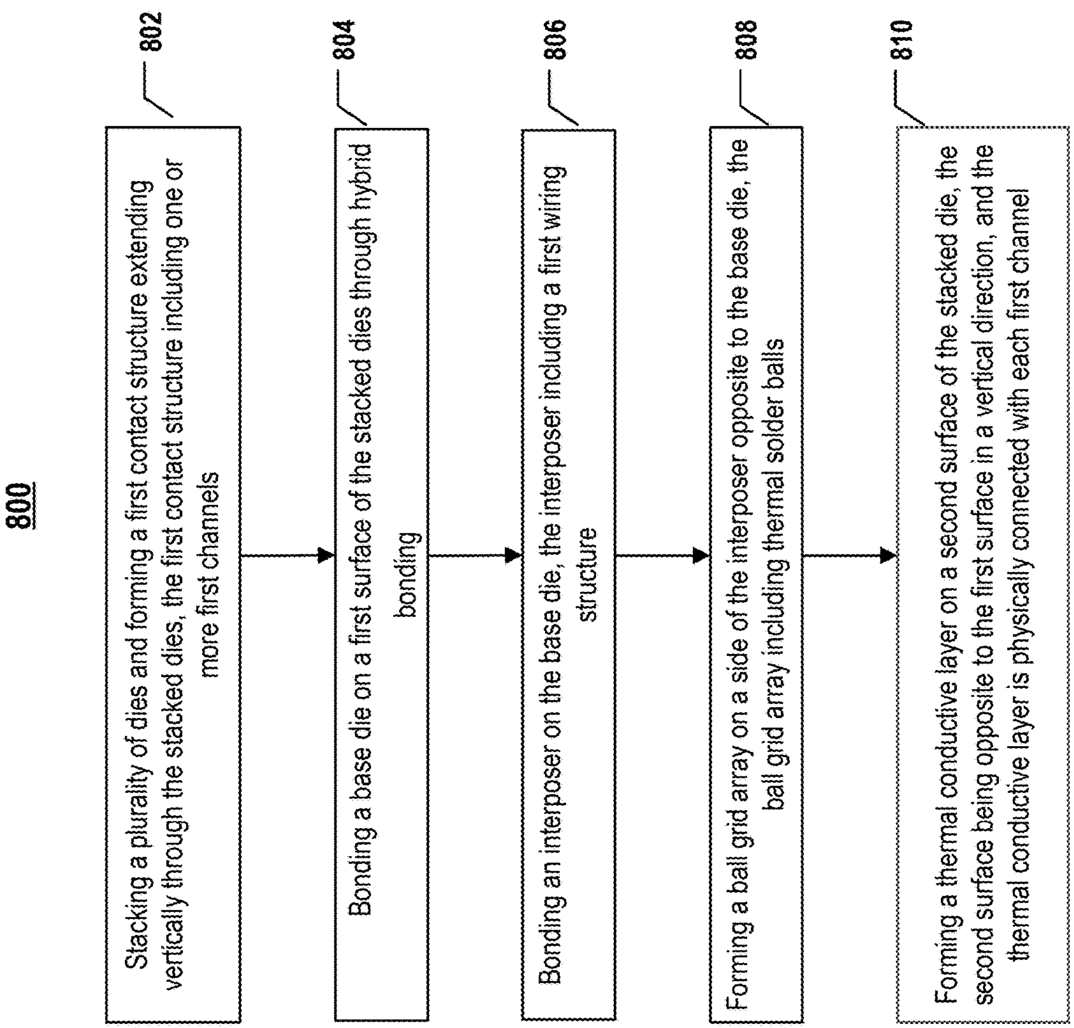
800
Stacking a plurality of dies and forming a first contact structure extending vertically through the stacked dies, the first contact structure including one or more first channels
802
Bonding a base die on a first surface of the stacked dies through hybrid bonding
804
Bonding an interposer on the base die, the interposer including a first wiring structure
806
Forming a ball grid array on a side of the interposer opposite to the base die, the ball grid array including thermal solder balls
808
Forming a thermal conductive layer on a second surface of the stacked die, the second surface being opposite to the first surface in a vertical direction, and the thermal conductive layer is physically connected with each first channel
810

FIG. 8

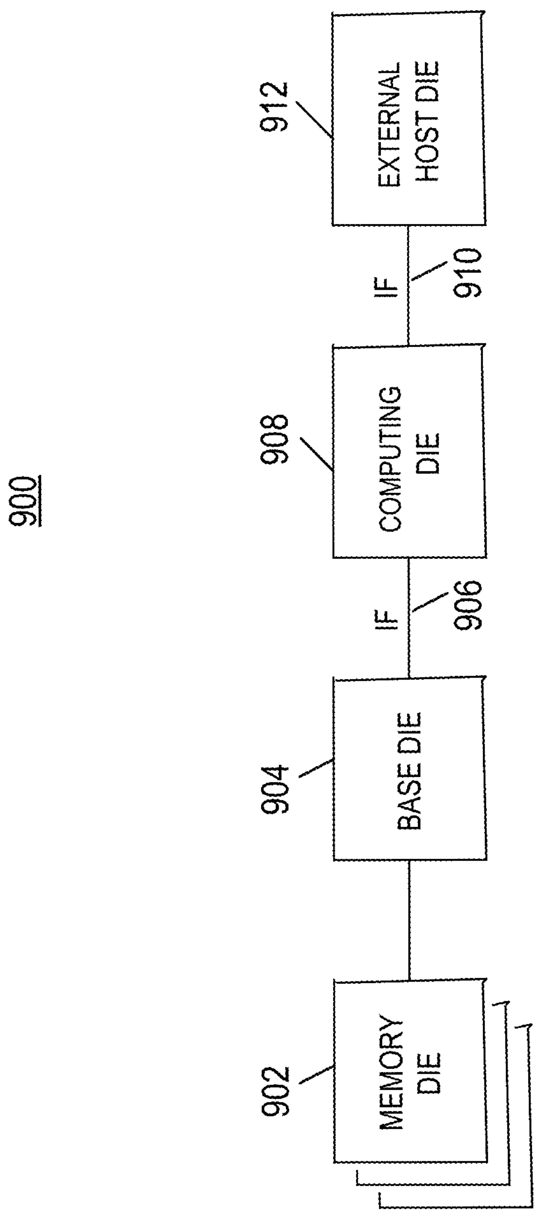
900
902
MEMORY DIE
904
BASE DIE
IF
906
908
COMPUTING DIE
IF
910
912
EXTERNAL HOST DIE

FIG. 9

SEMICONDUCTOR DEVICE AND FABRICATING METHOD THEREOF

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of International Application No. PCT/CN2024/081216, filed on Mar. 12, 2024, entitled "SEMICONDUCTOR DEVICE AND FABRICATING METHOD THEREOF," which claims the benefit of priority to International Application No. PCT/CN2024/078561, filed on Feb. 26, 2024, both of which are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

[0002] The present disclosure generally relates to the field of semiconductor technology, and more particularly, to packaging of semiconductor devices, and fabricating methods thereof.

BACKGROUND

[0003] As the power consumption of semiconductor chips continues increasing, the requirements for heat dissipation of packages become higher. However, the packaging structure has poor thermal conductivity on one hand, and the existing heat dissipation methods are mostly processed at the system level on the other hand. There is a demand for special heat dissipation inside the semiconductor packages.

SUMMARY

[0004] In one aspect, a semiconductor device is provided. The semiconductor device includes a plurality of dies stacked in a vertical direction; a thermal conductive layer deposited on a top surface of a topmost die; a base die located on a bottom surface of a bottommost die; and a first contact structure extending vertically through the plurality of dies. The first contact structure includes one or more first channels, each first channel having a first end being in contact with the base die and having a second end being in contact with the thermal conductive layer.

[0005] In some implementations, the semiconductor device further includes a second contact structure extending vertically inside the plurality of dies. The second contact structure includes one or more second channels, each second channel extending vertically and being in contact with the base die without contacting the thermal conductive layer.

[0006] In some implementations, the base die includes a wiring structure including a first wiring structure and a second wiring structure. The first wiring structure is configured to transport heat and the second wiring structure is configured to transmit electronic signals.

[0007] In some implementations, the semiconductor device further includes an interposer on a side of the base die opposite to the plurality of dies. The interposer includes a ball grid array on a side of the interposer opposite to the base dies.

[0008] In some implementations, the ball grid array includes one or more thermal solder balls in connection with the first wiring structure and one or more signal solder balls in connection with the second wiring structure.

[0009] In some implementations, each of the first channel and the second channel is a continuous structure along the vertical direction.

[0010] In some implementations, the second contact structure is surrounded by the first contact structure.

[0011] In some implementations, the first end of each first channel has a first lateral dimension, the second end of each first channel has a second lateral dimension, and the first lateral dimension is larger than the second lateral dimension.

[0012] In some implementations, each second channel has a first end in contact with the base die, the first end of each second channel has a third lateral dimension, and the third lateral dimension is smaller than a first lateral dimension of the first end of each first channel.

[0013] In some implementations, the first lateral dimension is 2-3 μm and the third lateral dimension is less than 2 μm.

[0014] In some implementations, the semiconductor device further includes a high operating power consumption region and a low operating power consumption region. A quantity of first channels in the high operating power consumption region is greater than a quantity of first channels in the low operating power consumption region.

[0015] In some implementations, a material of the thermal conductive layer has a thermal conductivity of no less than 20 W/mK.

[0016] In some implementations, the material of the thermal conductive layer is one of silicon nitride, aluminum oxide, and silicon carbide.

[0017] In some implementations, the plurality of dies includes at least one memory die, and the plurality of dies are stacked through direct bonding.

[0018] In another aspect, another semiconductor device is provided. The semiconductor device includes a plurality of dies stacked in a vertical direction; an interposer located on a bottom surface of a base die, where the interposer includes a wiring structure including a first wiring structure and a second wiring structure; and a first contact structure extending vertically through the plurality of dies. The first contact structure includes one or more first channels, each first channel having a first end being in contact with the interposer and having a second end being in contact with a top surface of a topmost die.

[0019] In some implementations, the semiconductor device includes a second contact structure extending vertically inside the plurality of dies. The second contact structure includes one or more second channels, each second channel extending vertically and being in contact with the interposer without contacting the top surface of the topmost die.

[0020] In some implementations, the semiconductor device includes a ball grid array on a side of the interposer opposite to the plurality of dies.

[0021] In some implementations, the ball grid array includes one or more thermal solder balls in contact with the first wiring structure, where the first wiring structure is configured to transport heat; and one or more signal solder balls in contact with the second wiring structure, where the second wiring structure is configured to transmit electronic signals.

[0022] In some implementations, each of the first channel and the second channel is a continuous structure along the vertical direction.

[0023] In some implementations, the second contact structure is surrounded by the first contact structure.

[0024] In some implementations, the first end of each first channel has a first lateral dimension, the second end of each first channel has a second lateral dimension, and the first lateral dimension is larger than the second lateral dimension.

[0025] In some implementations, each second channel has a first end in contact with the interposer, the first end of each second channel has a third lateral dimension, and the third lateral dimension is smaller than a first lateral dimension of the first end of each first channel.

[0026] In some implementations, the first lateral dimension is 2-3 μm and the third lateral dimension is less than 2 μm.

[0027] In some implementations, the semiconductor device includes a high operating power consumption region and a low operating power consumption region. A quantity of first channels in the high operating power consumption region is greater than a quantity of first channels in the low operating power consumption region.

[0028] In some implementations, a material of the first channels has a thermal conductivity of no less than 20 W/mK.

[0029] In some implementations, the material of the first channels is one of silicon nitride, aluminum oxide, and silicon carbide.

[0030] In some implementations, the plurality of dies include at least one memory die, and the plurality of dies are stacked through direct bonding.

[0031] In still another aspect, a method is provided. The method includes: stacking a plurality of dies in a vertical direction; depositing a sacrificial layer on a first surface of the stacked dies, and forming a plurality of first holes in the sacrificial layer; forming, using the plurality of first holes, a first contact structure extending vertically through the stacked dies, where the first contact structure includes one or more first channels; bonding a base die on the first surface of the stacked dies through hybrid bonding; bonding an interposer on the base die, where the interposer includes a wiring structure; connecting each first channel with the wiring structure; and forming a thermal conductive layer on a second surface of the stacked die, where the second surface is opposite to the first surface in the vertical direction, and the thermal conductive layer is physically connected with each first channel.

[0032] In some implementations, the method further includes: forming a plurality of second holes in the sacrificial layer; forming, using the plurality of second holes, a second contact structure extending vertically inside the stacked dies, where the second contact structure includes one or more second channels; and connecting each second channel with the interposer without physically contacting the thermal conductive layer.

[0033] In some implementations, the method further includes forming a ball grid array on a side of the interposer opposite to the stacked dies.

[0034] In some implementations, the wiring structure includes a first wiring structure configured to transport heat and a second wiring structure configured to transmit electronic signals, and forming the ball grid array includes: forming a plurality of thermal solder balls in contract with the first wiring structure; and forming a plurality of signal solder balls in contact with the second wiring structure.

[0035] In some implementations, forming the first contact structure includes forming each first channel continuously extending along the vertical direction; and forming the second contact structure includes forming each second channel continuously extending along the vertical direction.

[0036] In some implementations, forming the second contact structure includes forming the second contact structure surrounded by the first contact structure.

[0037] In some implementations, the method further includes forming each first channel with a first end having a first lateral dimension and a second end having a second lateral dimension. The first end of each first channel is in contact with the base die, the second end of each first channel is in contact with the thermal conductive layer, and the first lateral dimension is larger than the second lateral dimension.

[0038] In some implementations, the method further includes forming each second channel with a first end having a third lateral dimension. The first end of each second channel is contact with the base die and the third lateral dimension is smaller than the first lateral dimension.

[0039] In some implementations, forming the thermal conductive layer includes using a material having a thermal conductivity of no less than 20 W/mK.

[0040] In some implementations, forming the thermal conductive layer includes using one of silicon nitride, aluminum oxide, and silicon carbide.

[0041] In some implementations, stacking the plurality of dies includes stacking at least one memory die with other dies through direct bonding.

BRIEF DESCRIPTION OF THE DRAWINGS

[0042] The accompanying drawings, which are incorporated herein and form a part of the specification, illustrate implementations of the present disclosure and, together with the description, serve to explain the principles of the present disclosure and to enable a person skilled in the pertinent art to make and use the present disclosure.

[0043] FIG. 1A illustrates a schematic diagram in a perspective side view of a first exemplary semiconductor device, in accordance with some implementations of the present disclosure.

[0044] FIG. 1B illustrates a schematic diagram in a perspective side view of a second exemplary semiconductor device, in accordance with some implementations of the present disclosure.

[0045] FIG. 1C illustrates a schematic diagram in a perspective side view of a third exemplary semiconductor device, in accordance with some implementations of the present disclosure.

[0046] FIGS. 2A-2F illustrate a fabrication process for forming a first semiconductor device according to some aspects of the present disclosure.

[0047] FIGS. 3A-3F illustrate a fabrication process for forming a second semiconductor device according to some aspects of the present disclosure.

[0048] FIGS. 4A-4G illustrate a fabrication process for forming a third semiconductor device according to some aspects of the present disclosure.

[0049] FIGS. 5A-5B illustrate schematic diagrams in a planar view of an exemplary semiconductor device, in accordance with some other implementations of the present disclosure.

[0050] FIG. 6 illustrates a flow diagram of an exemplary method for forming a first semiconductor device, in accordance with some implementations of the present disclosure.

[0051] FIG. **7** illustrates a flow diagram of an exemplary method for forming a second semiconductor device, in accordance with some implementations of the present disclosure.

[0052] FIG. **8** illustrates a flow diagram of an exemplary method for forming a third semiconductor device, in accordance with some implementations of the present disclosure.

[0053] FIG. **9** illustrates a block diagram of an example system having one or more semiconductor devices, in accordance with some implementations of the present disclosure.

[0054] Implementations of the present disclosure will be described with reference to the accompanying drawings.

DETAILED DESCRIPTION

[0055] Although specific configurations and arrangements are discussed, it should be understood that this is done for illustrative purposes only. A person skilled in the pertinent art will recognize that other configurations and arrangements can be used without departing from the spirit and scope of the present disclosure. It will be apparent to a person skilled in the pertinent art that the present disclosure can also be employed in a variety of other applications.

[0056] It is noted that references in the specification to "one implementation," "an implementation," "an example implementation," "some implementations," etc., indicate that the implementation described can include a particular feature, structure, or characteristic, but every implementation can not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases do not necessarily refer to the same implementation. Further, when a particular feature, structure or characteristic is described in connection with an implementation, it would be within the knowledge of a person skilled in the pertinent art to affect such feature, structure, or characteristic in connection with other implementations whether or not explicitly described.

[0057] In general, terminology can be understood at least in part from usage in context. For example, the term "one or more" as used herein, depending at least in part upon context, can be used to describe any feature, structure, or characteristic in a singular sense or can be used to describe combinations of features, structures, or characteristics in a plural sense. Similarly, terms, such as "a," "an," or "the," again, can be understood to convey a singular usage or to convey a plural usage, depending at least in part upon context. In addition, the term "based on" can be understood as not necessarily intended to convey an exclusive set of factors and may instead, allow for existence of additional factors not necessarily expressly described, again, depending at least in part on context.

[0058] It should be readily understood that the meaning of "on," "above," and "over" in the present disclosure should be interpreted in the broadest manner such that "on" not only means "directly on" something, but also includes the meaning of "on" something with an intermediate feature or a layer therebetween. Moreover, "above" or "over" not only means "above" or "over" something, but can also include the meaning it is "above" or "over" something with no intermediate feature or layer therebetween (i.e., directly on something).

[0059] Further, spatially relative terms, such as "beneath," "below," "lower," "above," "upper," and the like, can be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or process step in addition to the orientation depicted in the figures. The apparatus can be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein can likewise be interpreted accordingly.

[0060] As used herein, the term "substrate" refers to a material onto which subsequent material layers are added. The substrate includes a "top" surface and a "bottom" surface. The front surface of the substrate is typically where a semiconductor device is formed, and therefore the semiconductor device is formed at a top side of the substrate unless stated otherwise. The bottom surface is opposite to the front surface and therefore a bottom side of the substrate is opposite to the top side of the substrate. The substrate itself can be patterned. Materials added on top of the substrate can be patterned or can remain unpatterned. Furthermore, the substrate can include a wide array of semiconductor materials, such as silicon, germanium, gallium arsenide, indium phosphide, etc. Alternatively, the substrate can be made from an electrically non-conductive material, such as a glass, a plastic, or a sapphire wafer.

[0061] As used herein, the term "layer" refers to a material portion including a region with a thickness. A layer has a top side and a bottom side where the bottom side of the layer is adjacent to the substrate and the top side is relatively away from the substrate. A layer can extend over the entirety of an underlying or overlying structure, or can have an extent less than the extent of an underlying or overlying structure. Further, a layer can be a region of a homogeneous or inhomogeneous continuous structure that has a thickness less than the thickness of the continuous structure. For example, a layer can be located between any set of horizontal planes between, or at, a top surface and a bottom surface of the continuous structure. A layer can extend horizontally, vertically, and/or along a tapered surface. A substrate can be a layer, can include one or more layers therein, and/or can have one or more layer thereupon, thereabove, and/or therebelow. A layer can include multiple layers. For example, an interconnect layer can include one or more conductive and contact layers (in which contacts, interconnect lines, and/or vertical interconnect accesses (VIAs) are formed) and one or more dielectric layers.

[0062] As used herein, the term "nominal/nominally" refers to a desired, or target, value of a characteristic or parameter for a component or a process step, set during the design phase of a product or a process, together with a range of values above and/or below the desired value. The range of values can be due to slight variations in manufacturing processes or tolerances. As used herein, the term "about" indicates the value of a given quantity that can vary based on a particular technology node associated with the subject semiconductor device. Based on the particular technology node, the term "about" can indicate a value of a given quantity that varies within, for example, 10-30% of the value (e.g., ±10%, ±20%, or ±30% of the value).

[0063] In the present disclosure, the term "horizontal/horizontally/lateral/laterally" means nominally parallel to a lateral surface of a substrate, and the term "vertical" or "vertically" means nominally perpendicular to the lateral surface of a substrate.

[0064] The present disclosure provides semiconductor devices with a high thermal conductivity. In the era of great computing power and artificial intelligence (AI), the von

Neumann architecture that combines high-bandwidth memory (HBM) with GPU remains the mainstream in the high-performance chip market. However, the high amount of heat generated by the stacked HBM is an urgent issue that needs to be addressed. In the traditional HBM architecture, as the number of stacked DRAM layers increases, the heat dissipation for different layers of DRAM varies, which can easily lead to slow heat dissipation for DRAMs located in the middle, causing a decline in performance. Because DRAM needs to continuously perform read and write operations, it tends to generate heat during operation. The stacked structure of HBM limits the heat dissipation of the DRAM located in the middle layers.

[0065] In the semiconductor devices provided by this disclosure, the heat generated during the operation of HBM can be rapidly dissipated to prevent localized overheating. The number of I/O channels can be significantly increased. Compared to some existing semiconductor devices, the disclosed semiconductor devices can achieve better internal heat dissipation inside stacked dies without increasing the chip area of the HBM.

[0066] FIG. **1**A shows a schematic diagram in a perspective side view of a first exemplary semiconductor device, in accordance with some implementations of the present disclosure. As shown in FIG. **1**A, semiconductor device **100**A can include an interposer **120**, a base die **140** on a first side of the interposer **120**, stacked dies **110** on a first side of the base die **140**, and a ball grid array (BGA) **130** on a second side of the interposer **120** opposite to the first side. It is noted that, the semiconductor device **100**A can further include any other suitable components that are not shown in FIG. **1**A. For example, the semiconductor device can further include a printed circuit board (PCB) on which the semiconductor device **100**A is mounted via the BGA **130**.

[0067] The interposer **120** can be any suitable semiconductor material having any suitable structure, such as a monocrystalline single-layer material, a polycrystalline silicon (polysilicon) single-layer material, a polysilicon and metal multi-layer material, etc.

[0068] The base die **140** can include wiring structure **144**, which includes first wiring structure **146** and second wiring structure **148** embedded in the base die. The first wiring structure **146** can include any suitable thermal conductive interconnection structures, such as conductive vias and patterned conductive layers, etc., and configured to transport heat. The second wiring structure **148** can include any suitable conductive interconnection structures, such as conductive channels, and configured to transmit electronic signals. In some implementations, the second wiring structure **148** can be configured to transmit both electronic signals and heat. The first wiring structure **146** and the second wiring structure **148** can be isolated from each other.

[0069] In some implementations, the first wiring structure **146** can include any suitable thermal conductive materials, such as copper (Cu), nickel (Ni), gold (Au), silver (Ag), platinum (Pt), cobalt (Co), titanium (Ti), chrome (Cr), zirconium (Zr), molybdenum (Mo), ruthenium (Ru), hafnium (Hf), tungsten (W), rhenium (Re), graphite, carbon black, combinations thereof, and/or other materials known to those skilled in the art. The second wiring structure **148** can comprise any suitable conductive materials, such as Ag, Cu, aluminum (Al), aluminum nitride, silicon carbide, W, graphite, zinc (Zn), combinations thereof, and/or other materials known to those skilled in the art. In some implementations, a ratio between a first wiring width of the first wiring structure **146** and a second wiring width of the second wiring structure **148** can be in a range between about 1.5:1 to about 2:1.

[0070] The stacked dies **110** can include a plurality of dies stacked vertically through direct bonding, which forms bonding between surfaces without using intermediate layers, such as solder or adhesives. The stacked dies can be attached to a first side of the base die **140** by an adhesive film (not shown). Exemplarily, the stacked dies **110** include die **102**, die **103**, die **104**, and die **105**. In some implementations, the stacked dies **110** can be any suitable semiconductor dies/die including one or more memory dies. The stacked dies **110** can include low power chips having maximum operation powers lower than a threshold power value. The stacked dies can include high power chips having maximum operation powers higher than the threshold power value. For example, the stacked dies **110** include at least one of a micro processing chip, a logic control chip, a power management chip, a driver chip, and an analog chip, and may also include at least one of a memory chip and a sensing chip. In some implementations, the adhesive film can be any suitable die attach film (DAF).

[0071] In some implementations, the semiconductor device includes a first contact structure **180** extending vertically in the stacked dies. The first contact structure can include one or more first channels **182**, and each first channel **182** is a continuous channel that has a first end in connection with the first wiring structure **146** and a second end in contact with a top surface of a topmost die of the stacked dies. The first end of each first channel **182** has a first lateral dimension (e.g., x-direction), the second end of each first channel **182** has a second lateral dimension (e.g., x-direction), and the first lateral dimension is larger than the second lateral dimension. In some implementations, the first lateral dimension is about 2-3 μm. In some implementations, the material of each first channel has a thermal conductive coefficient of no less than 20 W/mK. In some implementations, the material of each first channel is one of a metal, a ceramic material, or a silicon material. In some implementations, the material of each first channel can be one of silicon nitride, aluminum oxide, and silicon carbide.

[0072] In some implementations, the semiconductor device includes a second contact structure **185** extending vertically in the stacked dies. The second contact structure **185** can be sandwiched between two first contact structures **180** as shown in FIG. **5**A, or can be surrounded by the first contact structure **180** as shown in FIG. **5**B. The second contact structure **185** can include a plurality of second channels, and each second channel of second contact structure **185** is a continuous channel that has a first end in contact with the second wiring structure **148** and a second end inside the stacked dies without contacting the top surface of the topmost die of the stacked dies. The first end of each second channel of second contact structure **185** has a third lateral dimension (e.g., x-direction), and the third lateral dimension is smaller than the first lateral dimension of the first end of each first channel. In some implementations, the third lateral dimension is less than 2 μm. In some implementations, the plurality of second channels have different dimensions along the vertical direction. For example, second channel **185-1** has a first end in contact with the second wiring structure **148** and a second end inside the die **105** without contacting the top surface of die **105**; second channel **185-2** has a first end in contact with the second wiring structure **148** and a second end inside die **104** without contacting the top surface of die **104**; second channel **185-3** has a first end in contact with the second wiring structure **148** and a second end inside die **103** without contacting the top surface of die **103**; and second channel **185-4** has a first end in contact with the second wiring structure **148** and a second end inside die **102** without contacting the top surface of die **102**. In some implementations, the material of each second channel is one of a metal, a ceramic material, or a silicon material.

[0073] In some implementations, a thermal conductive layer **106** can cover the top surface of the topmost die, so that the thermal conductive layer **106** is in contact with each first channel **182**. The material of the thermal conductive layer **106** has a thermal conductive coefficient of no less than 20 W/mK. In some implementations, the material of the thermal conductive layer **106** is one of a metal, a ceramic material, or a silicon material. In some implementations, the material of each first channel can be one of silicon nitride, aluminum oxide, and silicon carbide.

[0074] In some implementations, the ball grid array (BGA) **130** can include a plurality of solder balls **132/135** attached on a second side of the interposer opposite to the first side. BGA **130** can include a plurality of thermal solder balls **132** configured to transport heat, and a plurality of signal solder balls **135** configured to transmit electronic signals.

[0075] In some implementations, the thermal solder balls **132** and the signal solder balls **135** can comprise same materials, and can be formed in a same process. For example, the thermal solder balls **132** and the signal solder balls **135** can include any suitable metal material, such as Aluminum (Al), Antimony (Sb), Arsenic (As), Bismuth (Bi), Cadmium (Cd), Co, Cu, Ni, Au, Ag, Indium (In), Iron (Fe), Lead (Pb), Phosphorus (P), Tin (Sn), Sulfur(S), Zinc (Zn), Germanium (Ge), etc., and any suitable alloy thereof. In some other implementations, the thermal solder balls **132** and the signal solder balls **135** can include different materials. For example, the thermal solder balls **132** can include a first material with a high thermal conductive coefficient, while the signal solder balls **135** can include a second material with a high electric conductive coefficient.

[0076] FIG. **1**B shows a schematic diagram in a perspective side view of a second exemplary semiconductor device, in accordance with some implementations of the present disclosure. As shown in FIG. **1**B, semiconductor device **100**B can include an interposer **120**, a base die **140** on a first side of the interposer **120**, stacked dies **110** on a first side of the base die **140**, and a ball grid array (BGA) **130** on a second side of the interposer **120** opposite to the first side. It is noted that, the semiconductor device **100**B can further include any other suitable components that are not shown in FIG. **1**B. For example, the semiconductor device can further include a printed circuit board (PCB) on which the semiconductor device **100**B is mounted via the BGA **130**.

[0077] The interposer **120** can be any suitable semiconductor material having any suitable structure, such as a monocrystalline single-layer material, a polycrystalline silicon (polysilicon) single-layer material, a polysilicon and metal multi-layer material, etc. The interposer **120** can include connecting structure **124**, which includes first connecting structure **126** and second connecting structure **128** embedded in the interposer **120**. The first connecting structure **126** can include any suitable thermal conductive interconnection structures, such as conductive vias and patterned conductive layers, etc., and is configured to transport heat. The second connecting structure **128** can include any suitable conductive interconnection structures, such as conductive channels, and is configured to transmit electronic signals. In some implementations, the second connecting structure **128** can be configured to transmit both electronic signals and heat. The first connecting structure **126** and the second connecting structure **128** can be isolated from each other.

[0078] In some implementations, the first connecting structure **126** can include any suitable thermal conductive materials, such as copper (Cu), nickel (Ni), gold (Au), silver (Ag), platinum (Pt), cobalt (Co), titanium (Ti), chrome (Cr), zirconium (Zr), molybdenum (Mo), ruthenium (Ru), hafnium (Hf), tungsten (W), rhenium (Re), graphite, carbon black, combinations thereof, and/or other materials known to those skilled in the art. The second connecting structure **128** can comprise any suitable conductive materials, such as Ag, Cu, aluminum (Al), aluminum nitride, silicon carbide, W, graphite, zinc (Zn), combinations thereof, and/or other materials known to those skilled in the art.

[0079] The base die **140** can include wiring structure **144**, which includes first wiring structure **146** and second wiring structure **148** embedded in the base die. The first wiring structure **146** can include any suitable thermal conductive interconnection structures, such as conductive vias and patterned conductive layers, etc., and configured to transport heat. The second wiring structure **148** can include any suitable conductive interconnection structures, such as conductive channels, and configured to transmit electronic signals. In some implementations, the second wiring structure **148** can be configured to transmit both electronic signals and heat. In some implementations, the first wiring structure **146** is physically connected with the first connecting structure **126**, and the second wiring structure **148** is physically connected with the second connecting structure **128**. The first wiring structure **146** and the second wiring structure **148** can be isolated from each other.

[0080] In some implementations, the first wiring structure **146** can include any suitable thermal conductive materials, such as copper (Cu), nickel (Ni), gold (Au), silver (Ag), platinum (Pt), cobalt (Co), titanium (Ti), chrome (Cr), zirconium (Zr), molybdenum (Mo), ruthenium (Ru), hafnium (Hf), tungsten (W), rhenium (Re), graphite, carbon black, combinations thereof, and/or other materials known to those skilled in the art. The second wiring structure **148** can comprise any suitable conductive materials, such as Ag, Cu, aluminum (Al), aluminum nitride, silicon carbide, W, graphite, zinc (Zn), combinations thereof, and/or other materials known to those skilled in the art. In some implementations, a ratio between a first wiring width of the first wiring structure **146** and a second wiring width of the second wiring structure **148** can be in a range between about 1.5:1 to about 2:1.

[0081] The stacked dies **110** can include a plurality of dies stacked vertically through direct bonding, which forms bonding between surfaces without using intermediate layers, such as solder or adhesives. The stacked dies are attached to a first side of the base die **140** by an adhesive film (not shown). Exemplarily, the stacked dies **110** include die **102**, die **103**, die **104**, and die **105**. In some implementations, the stacked dies **110** can be any suitable semiconductor dies/die including one or more memory dies. The stacked dies **110**

can include low power chips having maximum operation powers lower than a threshold power value. The stacked dies can include high power chips having maximum operation powers higher than the threshold power value. For example, the stacked dies **110** include at least one of a micro processing chip, a logic control chip, a power management chip, a driver chip, and an analog chip, and may also include at least one of a memory chip and a sensing chip. In some implementations, the adhesive film can be any suitable die attach film (DAF).

[0082] In some implementations, the semiconductor device includes a first contact structure **180** extending vertically in the stacked dies. The first contact structure can include one or more first channels **182**, and each first channel **182** is a continuous channel that has a first end in connection with the first wiring structure **146** and a second end in contact with a top surface of a topmost die of the stacked dies. The first end of each first channel **182** has a first lateral dimension (e.g., x-direction), the second end of each first channel **182** has a second lateral dimension (e.g., x-direction), and the first lateral dimension is larger than the second lateral dimension. In some implementations, the first lateral dimension is about 2-3 μm. In some implementations, the material of each first channel has a thermal conductive coefficient of no less than 20 W/mK. In some implementations, the material of each first channel is one of a metal, a ceramic material, or a silicon material. In some implementations, the material of each first channel can be one of silicon nitride, aluminum oxide, and silicon carbide.

[0083] In some implementations, the semiconductor device includes a second contact structure **185** extending vertically in the stacked dies. The second contact structure **185** can be sandwiched between two first contact structures **180** as shown in FIG. **5**A, or can be surrounded by the first contact structure **180** as shown in FIG. **5**B. The second contact structure **185** can include a plurality of second channels, and each second channel of second contact structure **185** is a continuous channel that has a first end in contact with the second wiring structure **148** and a second end inside the stacked dies without contacting the top surface of the topmost die of the stacked dies. The first end of each second channel **185** has a third lateral dimension (e.g., x-direction), and the third lateral dimension is smaller than the first lateral dimension of the first end of each first channel. In some implementations, the third lateral dimension is less than 2 μm. In some implementations, the plurality of second channels have different dimensions along the vertical direction. For example, second channel **185-1** has a first end in contact with the second wiring structure **148** and a second end inside the die **105** without contacting the top surface of die **105**; second channel **185-2** has a first end in contact with the second wiring structure **148** and a second end inside die **104** without contacting the top surface of die **104**; second channel **185-3** has a first end in contact with the second wiring structure **148** and a second end inside die **103** without contacting the top surface of die **103**; and second channel **185-4** has a first end in contact with the second wiring structure **148** and a second end inside die **102** without contacting the top surface of die **102**. In some implementations, the material of each second channel is one of a metal, a ceramic material, or a silicon material.

[0084] In some implementations, the ball grid array (BGA) **130** can include a plurality of solder balls **132/135** attached on a second side of the interposer opposite to the first side. BGA **130** can include a plurality of thermal solder balls **132** in contact with the first connecting structure **126**, and a plurality of signal solder ball **135** in contact with the second connecting structure **128**. That is, the thermal solder balls **132** can be coupled with the first contact structure **180** via the first wiring structure **146** and first connecting structure **126**, and are configured to dissipate heat in the stacked dies. The signal solder balls **135** can be coupled with the second contact structure **185** via the second wiring structure **148** and the second connecting structure **128**, and are configured to transmit electronic signals between the stacked dies and the PCB.

[0085] In some implementations, the thermal solder balls **132** and the signal solder balls **135** can comprise same materials, and can be formed in a same process. For example, the thermal solder balls **132** and the signal solder balls **135** can include any suitable metal material, such as Aluminum (Al), Antimony (Sb), Arsenic (As), Bismuth (Bi), Cadmium (Cd), Co, Cu, Ni, Au, Ag, Indium (In), Iron (Fe), Lead (Pb), Phosphorus (P), Tin (Sn), Sulfur(S), Zinc (Zn), Germanium (Ge), etc., and any suitable alloy thereof. In some other implementations, the thermal solder balls **132** and the signal solder balls **135** can include different materials. For example, the thermal solder balls **132** can include a first material with a high thermal conductive coefficient, while the signal solder balls **135** can include a second material with a high electric conductive coefficient.

[0086] FIG. **1**C shows a schematic diagram in a perspective side view of a third exemplary semiconductor device, in accordance with some implementations of the present disclosure. As shown in FIG. **1**C, semiconductor device **100**C can include an interposer **120**, a base die **140** on a first side of the interposer **120**, stacked dies **110** on a first side of the base die **140**, and a ball grid array (BGA) **130** on a second side of the interposer **120** opposite to the first side. It is noted that, the semiconductor device **100**C can further include any other suitable components that are not shown in FIG. **1**C. For example, the semiconductor device can further include a printed circuit board (PCB) on which the semiconductor device **100**C is mounted via the BGA **130**.

[0087] The interposer **120** can be any suitable semiconductor material having any suitable structure, such as a monocrystalline single-layer material, a polycrystalline silicon (polysilicon) single-layer material, a polysilicon and metal multi-layer material, etc. The interposer **120** can include connecting structure **124**, which includes first connecting structure **126** and second connecting structure **128** embedded in the interposer **120**. The first connecting structure **126** can include any suitable thermal conductive interconnection structures, such as conductive vias and patterned conductive layers, etc., and is configured to transport heat. The second connecting structure **128** can include any suitable conductive interconnection structures, such as conductive channels, and is configured to transmit electronic signals. In some implementations, the second connecting structure **128** can be configured to transmit both electronic signals and heat. The first connecting structure **126** and the second connecting structure **128** can be isolated from each other.

[0088] In some implementations, the first connecting structure **126** can include any suitable thermal conductive materials, such as copper (Cu), nickel (Ni), gold (Au), silver (Ag), platinum (Pt), cobalt (Co), titanium (Ti), chrome (Cr), zirconium (Zr), molybdenum (Mo), ruthenium (Ru), hafnium (Hf), tungsten (W), rhenium (Re), graphite, carbon black, combinations thereof, and/or other materials known to those skilled in the art. The second connecting structure **128** can comprise any suitable conductive materials, such as Ag, Cu, aluminum (Al), aluminum nitride, silicon carbide, W, graphite, zinc (Zn), combinations thereof, and/or other materials known to those skilled in the art.

[0089] The base die **140** can include wiring structure **144**, which includes first wiring structure **146** and second wiring structure **148** embedded in the base die. The first wiring structure **146** can include any suitable thermal conductive interconnection structures, such as conductive vias and patterned conductive layers, etc., and configured to transport heat. The second wiring structure **148** can include any suitable conductive interconnection structures, such as conductive channels, and configured to transmit electronic signals. In some implementations, the second wiring structure **148** can be configured to transmit both electronic signals and heat. In some implementations, the first wiring structure **146** is physically connected with the first connecting structure **126**, and the second wiring structure **148** is physically connected with the second connecting structure **128**. The first wiring structure **146** and the second wiring structure **148** can be isolated from each other.

[0090] In some implementations, the first wiring structure **146** can include any suitable thermal conductive materials, such as copper (Cu), nickel (Ni), gold (Au), silver (Ag), platinum (Pt), cobalt (Co), titanium (Ti), chrome (Cr), zirconium (Zr), molybdenum (Mo), ruthenium (Ru), hafnium (Hf), tungsten (W), rhenium (Re), graphite, carbon black, combinations thereof, and/or other materials known to those skilled in the art. The second wiring structure **148** can comprise any suitable conductive materials, such as Ag, Cu, aluminum (Al), aluminum nitride, silicon carbide, W, graphite, zinc (Zn), combinations thereof, and/or other materials known to those skilled in the art. In some implementations, a ratio between a first wiring width of the first wiring structure **146** and a second wiring width of the second wiring structure **148** can be in a range between about 1.5:1 to about 2:1.

[0091] The stacked dies **110** can include a plurality of dies stacked vertically through direct bonding, which forms bonding between surfaces without using intermediate layers, such as solder or adhesives. The stacked dies **110** can be attached to a first side of the base die **140** by an adhesive film (not shown). Exemplarily, the stacked dies **110** include die **102**, die **103**, die **104**, and die **105**. In some implementations, the stacked dies **110** can be any suitable semiconductor dies/die including one or more memory dies. The stacked dies can include low power chips having maximum operation powers lower than a threshold power value. The stacked dies can include high power chips having maximum operation powers higher than the threshold power value. For example, the stacked dies **110** include at least one of a micro processing chip, a logic control chip, a power management chip, a driver chip, and an analog chip, and may also include at least one of a memory chip and a sensing chip. In some implementations, the adhesive film can be any suitable die attach film (DAF).

[0092] In some implementations, the semiconductor device includes a first contact structure **180** extending vertically in the stacked dies. The first contact structure can include one or more first channels **182**, and each first channel **182** is a continuous channel that has a first end in connection with the first wiring structure **146** and a second end in contact with a top surface of a topmost die of the stacked dies. The first end of each first channel **182** has a first lateral dimension (e.g., x-direction), the second end of each first channel **182** has a second lateral dimension (e.g., x-direction), and the first lateral dimension is larger than the second lateral dimension. In some implementations, the first lateral dimension is about 2-3 μm. In some implementations, the material of each first channel has a thermal conductive coefficient of no less than 20 W/mK. In some implementations, the material of each first channel is one of a metal, a ceramic material, or a silicon material. In some implementations, the material of each first channel can be one of silicon nitride, aluminum oxide, and silicon carbide.

[0093] In some implementations, the semiconductor device includes a second contact structure **185** extending vertically in the stacked dies. The second contact structure **185** can be sandwiched between two first contact structures **180** as shown in FIG. **5**A, or can be surrounded by the first contact structure **180** as shown in FIG. **5**B. The second contact structure **185** can include a plurality of second channels, and each second channel of second contact structure **185** is a continuous channel that has a first end in contact with the second wiring structure **148** and a second end inside the stacked dies without contacting the top surface of the topmost die of the stacked dies. The first end of each second channel **185** has a third lateral dimension (e.g., x-direction), and the third lateral dimension is smaller than the first lateral dimension of the first end of each first channel. In some implementations, the third lateral dimension is less than 2 μm. In some implementations, the plurality of second channels have different dimensions along the vertical direction. For example, second channel **185-1** has a first end in contact with the second wiring structure **148** and a second end inside the die **105** without contacting the top surface of die **105**; second channel **185-2** has a first end in contact with the second wiring structure **148** and a second end inside die **104** without contacting the top surface of die **104**; second channel **185-3** has a first end in contact with the second wiring structure **148** and a second end inside die **103** without contacting the top surface of die **103**; and second channel **185-4** has a first end in contact with the second wiring structure **148** and a second end inside die **102** without contacting the top surface of die **102**. In some implementations, the material of each second channel is one of a metal, a ceramic material, or a silicon material.

[0094] In some implementations, a thermal conductive layer **106** can cover the top surface of the topmost die, so that the thermal conductive layer **106** is in contact with each first channel. A material of the thermal conductive layer **106** has a thermal conductive coefficient no less than 20 W/mK. In some implementations, the material of the thermal conductive layer **106** is one of a metal, a ceramic material, or a silicon material.

[0095] In some implementations, the ball grid array (BGA) **130** can include a plurality of solder balls **132**/**135** attached on a second side of the interposer opposite to the first side. BGA **130** can include a plurality of thermal solder balls **132** in contact with the first connecting structure **126**, and a plurality of signal solder ball **135** in contact with the second connecting structure **128**. That is, the thermal solder balls **132** can be coupled with the first contact structure **180** via the first wiring structure **146** and first connecting structure **126**, and are configured to dissipate heat in the stacked dies. The signal solder balls **135** can be coupled with the second contact structure **185** via the second wiring structure **148** and the second connecting structure **128**, and are configured to transmit electronic signals between the stacked dies and the PCB.

[0096] In some implementations, the thermal solder balls **132** and the signal solder balls **135** can comprise same materials, and can be formed in a same process. For example, the thermal solder balls **132** and the signal solder balls **135** can include any suitable metal material, such as Aluminum (Al), Antimony (Sb), Arsenic (As), Bismuth (Bi), Cadmium (Cd), Co, Cu, Ni, Au, Ag, Indium (In), Iron (Fe), Lead (Pb), Phosphorus (P), Tin (Sn), Sulfur(S), Zinc (Zn), Germanium (Ge), etc., and any suitable alloy thereof. In some other implementations, the thermal solder balls **132** and the signal solder balls **135** can include different materials. For example, the thermal solder balls **132** can include a first material with a high thermal conductive coefficient, while the signal solder balls **135** can include a second material with a high electric conductive coefficient.

[0097] Referring to FIG. **6**, a flow diagram of an exemplary method for forming a first semiconductor device is illustrated in accordance with some implementations of the present disclosure. It should be understood that the operations and/or steps shown in FIG. **6** are not exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. FIGS. **2A-2F** illustrate schematic diagrams in a perspective side view of an exemplary semiconductor device at certain fabricating steps of the method shown in FIG. **6**, in accordance with some implementations of the present disclosure.

[0098] As shown in FIG. **6**, the method **600** starts at operation **602**, in which a plurality of dies are stacked along a vertical direction. A first contact structure is formed in the stacked dies. The first contact structure includes a plurality of first channels, each extending vertically through the stacked dies. FIGS. **2A-2C** illustrate schematic diagrams in a perspective side view of exemplary semiconductor devices during operation **602** of the method **600** shown in FIG. **6**, in accordance with some implementations of the present disclosure.

[0099] As shown in FIG. **2A**, in some implementations, a plurality of dies are stacked through direct bonding, which forms bonding between surfaces without using intermediate layers, such as solder or adhesives, in a vertical direction on a carrier wafer **215**. Exemplarily, as shown in FIG. **2A**, die **202**, die **203**, die **204**, and die **205** are stacked vertically through direct bonding, which forms bonding between surfaces without using intermediate layers, such as solder or adhesives, on the carrier wafer **215** to form stacked dies **210**. In some implementations, the stacked dies **210** can be any suitable semiconductor dies/die including one or more memory dies. The stacked dies **210** can include low power chips having maximum operation powers lower than a threshold power value. The stacked dies **210** can include high power chips having maximum operation powers higher than the threshold power value. For example, the stacked dies **210** include at least one of a micro processing chip, a logic control chip, a power management chip, a driver chip, and an analog chip, and may also include at least one of a memory chip and a sensing chip.

[0100] As shown in FIG. **2B**, in some implementations, a sacrificial layer **212** is deposited on a first surface of the stacked dies, and a plurality of first holes **270** are formed in the sacrificial layer **212** and vertically extend through the stacked dies **210**. In some implementations, the carrier wafer **215** is removed before forming the sacrificial layer **212**. In some implementations, the carrier wafer **215** is used as the sacrificial layer **212** and the plurality of first holes **270** are formed in the carrier wafer **215**. A lithography process can be performed to pattern the first holes using an etch mask (e.g., a photoresist mask and/or a hard mask), and one or more dry etching and/or wet etching processes, such as RIE, can be performed to etch first holes in the sacrificial layer **212**. Thus, the first holes **270** extending vertically in the sacrificial layer **212** can be formed.

[0101] As shown in FIG. **2B**, in some implementations, a plurality of second holes **272** are formed in the sacrificial layer **212** and vertically extend inside the stacked dies **210**. A lithography process can be performed to pattern the second holes using an etch mask (e.g., a photoresist mask and/or a hard mask), and one or more dry etching and/or wet etching processes, such as RIE, can be performed to etch second holes in the sacrificial layer **212**. Thus, second holes **272** extending vertically in the sacrificial layer **212** can be formed. In some implementations, the plurality of second holes **272** have different dimensions along the vertical direction. The dimensions of the second holes along the vertical direction can be controlled by etching each second hole with different time durations or by back-filling the second holes to achieve a certain vertical dimension. In some implementations, forming the second holes **272** can be performed in a same process of forming the first holes **270**. A lateral dimension (e.g., the x-direction) of each first hole **270** is larger than a lateral dimension (e.g., the x-direction) of each second hole **272**. In some implementations, the second holes **272** can be sandwiched between two sets of first holes **270**, or the second holes **272** can be surrounded by the plurality of first holes **270**.

[0102] As shown in FIG. **2C**, in some embodiments, a first contact structure **280** including a plurality of first channels **282** is formed inside the plurality of first holes **270**, for example, by depositing a metal, a ceramic material, or a silicon material, to fill first holes **270** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Each first channel **282** is a continuous channel that extends through the stacked dies. That is, each first channel **282** has a first end in contact with the bottom surface of the bottommost die and a second end in contact with the top surface of the topmost die of the stacked dies. The first end of each first channel **282** has a first lateral dimension (e.g., x-direction), the second end of each first channel **282** has a second lateral dimension (e.g., x-direction), and the first lateral dimension is larger than the second lateral dimension. In some implementations, the first lateral dimension of each first channel **282** is about 2-3 μm. The material of each first channel **282** has a thermal conductive coefficient of no less than 20 W/mK. In some implementations, the material of each first channel is one of a metal, a ceramic material, or a silicon material. In some implementations, the material of each first channel can be one of silicon nitride, aluminum oxide, and silicon carbide. A planarization process, such as CMP, is performed to remove excess first channel **282** that is deposited beyond the top surface of the topmost die.

[0103] In some implementations, as shown in FIG. **2C**, a second contact structure **285** including a plurality of second channels can be formed inside the plurality of second holes

272, for example, by depositing a metal, a ceramic material, or a silicon material, to fill second holes **272** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Each second channel is a continuous channel that extends vertically inside the stack dies. That is, each second channel has a first end in contact with the bottom surface of the bottommost die and a second end inside the stacked dies without contacting the top surface of the topmost die of the stacked dies. The first end of each second channel of second contact structure **285** has a third lateral dimension (e.g., x-direction), and the third lateral dimension is smaller than the first lateral dimension of the first end of each first channel **282**. In some implementations, the third lateral dimension is less than 2 μm. In some implementations, the plurality of second channels have different dimensions along the vertical direction. For example, second channel **285-1** has a first end in contact with the bottom surface of die **202** and a second end inside the die **202** without contacting the top surface of die **202**; second channel **285-2** has a first end in contact with the bottom surface of die **202** and a second end inside die **203** without contacting the top surface of die **203**; second channel **285-3** has a first end in contact with the bottom surface of die **202** and a second end inside die **203** without contacting the top surface of die **203**; and second channel **285-4** has a first end in contact with the top surface of die **202** and a second end inside die **205** without contacting the top surface of die **205**. In some implementations, the material of each second channel is one of a metal, a ceramic material, or a silicon material.

[0104] In some implementations, the first contact structure **280** and the second contact structure **285** are formed in a same process; or the first contact structure **280** and the second contact structure **285** can be formed in different processes. The first contact structure **280** and the second contact structure **285** can be made from a same material; or the first contact structure **280** and the second contact structure **285** can be made from different materials. In some implementations, the first channel **282** can include any suitable thermal conductive materials, such as copper (Cu), nickel (Ni), gold (Au), silver (Ag), platinum (Pt), cobalt (Co), titanium (Ti), chrome (Cr), zirconium (Zr), molybdenum (Mo), ruthenium (Ru), hafnium (Hf), tungsten (W), rhenium (Re), graphite, carbon black, combinations thereof, and/or other materials known to those skilled in the art. The second channel **285-1** can comprise any suitable conductive materials, such as Ag, Cu, aluminum (Al), aluminum nitride, silicon carbide, W, graphite, zinc (Zn), combinations thereof, and/or other materials known to those skilled in the art. In some implementations, a ratio between a first width of the first channel **282** and a second width of the second channel **285-1** can be in a range between about 1.5:1 to about 2:1.

[0105] The first contact structure **280** and the second contact structure **285** are then polished such that the surface of the first contact structure **280** and the surface of the second contact structure **285** are flush with the bottom surface of die **202**. In some implementations, a chemical-mechanical polishing (CMP) is performed to polish the first contact structure **280** and the second contact structure **285**.

[0106] Referring back to FIG. **6**, the method **600** can proceed to operation **604**, in which a base die is bonded with the stacked dies. FIG. **2**D illustrates a schematic diagram in a perspective side view of an exemplary semiconductor device after operation **604** of the method **600** shown in FIG. **6**, in accordance with some implementations of the present disclosure.

[0107] As shown in FIG. **2**D, a base die **240** can be bonded with a first surface (bottom surface) of the stacked dies **210** through hybrid bonding or direct bonding, such that a surface of the base die **240** is in contact with the first end of each first channel **282** of the first contact structure **280** and in contact with the first end of each second channel of second contact structure **285**. The base die **240** can be bonded with the stacked dies **210** through hybrid bonding or direct bonding, which is not limited herein. The base die **240** can be provided with wiring structure **244**, which includes first wiring structure **246** and second wiring structure **248** embedded in the base die. The first wiring structure **246** can include any suitable thermal conductive interconnection structures, such as conductive vias and patterned conductive layers, etc., and is configured to transport heat. The first wiring structure **246** is physically connected with the first contact structure **280**. For example, as shown in FIG. **2**D, the first wiring structure is physically connected with the first channel **282**. The second wiring structure **248** is physically connected with the second contact structure **285**. For example, as shown in FIG. **2**D, the second wiring structure is physically connected with the second channel **285-1**. The second wiring structure **248** can include any suitable conductive interconnection structures, such as conductive channels, and configured to transmit electronic signals. In some implementations, the second wiring structure **248** can be configured to transmit both electronic signals and heat. The first wiring structure **246** and the second wiring structure **248** can be isolated from each other.

[0108] In some implementations, the first wiring structure **246** can include any suitable thermal conductive materials, such as copper (Cu), nickel (Ni), gold (Au), silver (Ag), platinum (Pt), cobalt (Co), titanium (Ti), chrome (Cr), zirconium (Zr), molybdenum (Mo), ruthenium (Ru), hafnium (Hf), tungsten (W), rhenium (Re), graphite, carbon black, combinations thereof, and/or other materials known to those skilled in the art. The second wiring structure **248** can comprise any suitable conductive materials, such as Ag, Cu, aluminum (Al), aluminum nitride, silicon carbide, W, graphite, zinc (Zn), combinations thereof, and/or other materials known to those skilled in the art. In some implementations, a ratio between a first wiring width of the first wiring structure **246** and a second wiring width of the second wiring structure **248** can be in a range between about 1.5:1 to about 2:1.

[0109] Referring back to FIG. **6**, the method **600** can proceed to operation **606**, in which an interposer is bonded with the base die. FIG. **2**E illustrates a schematic diagram in a perspective side view of an exemplary semiconductor device after operation **606** of the method **600** shown in FIG. **6**, in accordance with some implementations of the present disclosure.

[0110] As shown in FIG. **2**E, an interposer **220** can be bonded with a surface (bottom surface) of the base die **240**, such that a surface of the interposer **220** is in contact with the wiring structure **244**. The interposer **220** can be bonded with the base die **240** through hybrid bonding or direct bonding, which is not limited herein. The interposer **220** can be any suitable semiconductor material having any suitable structure, such as a monocrystalline single-layer material, a polycrystalline silicon (polysilicon) single-layer material, a polysilicon and metal multi-layer material, etc. In some implementations, the interposer **220** is provided with a ball grid array (BGA) **230**.

[0111] Referring back to FIG. **6**, the method **600** can proceed to operation **608**, in which a thermal conductive layer is formed on a second surface (top surface) of the stacked dies. The second surface of the stacked dies is opposite to the first surface of the stacked dies in the vertical direction. FIG. **2**F illustrates a schematic diagram in a perspective side view of an exemplary semiconductor device after operation **608** of the method **600** shown in FIG. **6**, in accordance with some implementations of the present disclosure.

[0112] As shown in FIG. **2**F, a thermal conductive layer **206** can be formed by depositing a thermal conductive material to cover the top surface of the topmost die of the stacked dies **210**, so that the thermal conductive layer **206** is in contact with each first channel **282**. The material of the thermal conductive layer **206** has a thermal conductive coefficient of no less than 20 W/mK. In some implementations, the material of the thermal conductive layer **206** is one of a metal, a ceramic material, or a silicon material. In some implementations, the material of each first channel can be one of silicon nitride, aluminum oxide, and silicon carbide.

[0113] Referring to FIG. **7**, a flow diagram of an exemplary method for forming a second semiconductor device is illustrated in accordance with some implementations of the present disclosure. It should be understood that the operations and/or steps shown in FIG. **7** are not exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. FIGS. **3**A-**3**F illustrate schematic diagrams in a perspective side view of an exemplary semiconductor device at certain fabricating steps of the method shown in FIG. **7**, in accordance with some implementations of the present disclosure.

[0114] As shown in FIG. **7**, the method **700** starts at operation **702**, in which a plurality of dies are stacked along a vertical direction. A first contact structure is formed in the stacked dies. The first contact structure includes a plurality of first channels, each extending vertically through the stacked dies. FIGS. **3**A-**3**C illustrate schematic diagrams in a perspective side view of exemplary semiconductor devices during operation **702** of the method **700** shown in FIG. **7**, in accordance with some implementations of the present disclosure.

[0115] As shown in FIG. **3**A, in some implementations, a plurality of dies are stacked through direct bonding, which forms bonding between surfaces without using intermediate layers, such as solder or adhesives, in a vertical direction on a carrier wafer **315**. Exemplarily, as shown in FIG. **3**A, die **302**, die **303**, die **304**, and die **305** are stacked vertically on the carrier wafer **315** to form stacked dies **310**. In some implementations, the stacked dies **310** can be any suitable semiconductor dies/die including one or more memory dies. The stacked dies **310** can include low power chips having maximum operation powers lower than a threshold power value. The stacked dies **310** can include high power chips having maximum operation powers higher than the threshold power value. For example, the stacked dies **310** include at least one of a micro processing chip, a logic control chip, a power management chip, a driver chip, and an analog chip, and may also include at least one of a memory chip and a sensing chip.

[0116] As shown in FIG. **3**B, in some implementations, a sacrificial layer **312** is deposited on a first surface of the stacked dies **310**, and a plurality of first holes **370** are formed in the sacrificial layer **312** and vertically extend through the stacked dies **310**. In some implementations, the carrier wafer **315** is removed before forming the sacrificial layer **312**. In some implementations, the carrier wafer **315** is used as the sacrificial layer **312** and the plurality of first holes **270** are formed in the carrier wafer **315**. A lithography process can be performed to pattern the first holes using an etch mask (e.g., a photoresist mask and/or a hard mask), and one or more dry etching and/or wet etching processes, such as RIE, can be performed to etch first holes **370** in the sacrificial layer **312**. Thus, first holes **370** extending vertically in the sacrificial layer **312** can be formed.

[0117] As shown in FIG. **3**B, in some implementations, a plurality of second holes **372** are formed in the sacrificial layer **312** and vertically extend inside the stacked dies **310**. A lithography process can be performed to pattern the second holes **372** using an etch mask (e.g., a photoresist mask and/or a hard mask), and one or more dry etching and/or wet etching processes, such as RIE, can be performed to etch second holes **372** in the sacrificial layer **312**. Thus, second holes **372** extending vertically in the sacrificial layer **312** can be formed. In some implementations, the plurality of second holes **372** have different dimensions along the vertical direction. The dimensions of the second holes **372** along the vertical direction can be controlled by etching each second hole **372** with different time durations or by backfilling the second holes **372** to achieve a certain vertical dimension. In some implementations, forming the second holes **372** can be performed in a same process of forming the first holes **370**. A lateral dimension (e.g., the x-direction) of each first hole **370** is larger than a lateral dimension (e.g., the x-direction) of each second hole **372**. In some implementations, the second holes **372** can be sandwiched between two sets of first holes **370**, or the second holes **372** can be surrounded by the plurality of first holes **370**.

[0118] As shown in FIG. **3**C, in some embodiments, a first contact structure **380** including a plurality of first channels **382** is formed inside the plurality of first holes **370**, for example, by depositing a metal, a ceramic material, or a silicon material, to fill first holes **370** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Each first channel **382** is a continuous channel that extends through the stacked dies **310**. That is, each first channel **382** has a first end in contact with the bottom surface of the bottommost die and a second end in contact with the top surface of the topmost die of the stacked dies **310**. The first end of each first channel **382** has a first lateral dimension (e.g., x-direction), the second end of each first channel **382** has a second lateral dimension (e.g., x-direction), and the first lateral dimension is larger than the second lateral dimension. In some implementations, the first lateral dimension is about 2-3 μm. The material of each first channel **382** has a thermal conductive coefficient of no less than 20 W/mK. In some implementations, the material of each first channel **382** is one of a metal, a ceramic material, or a silicon material. In some implementations, the material of each first channel **382** can be one of silicon nitride, aluminum oxide, and silicon carbide. A planarization process, such as CMP, can be performed to remove excess first channel **382** that is deposited beyond the top surface of the topmost die.

[0119] In some implementations, as shown in FIG. 3C, a second contact structure **385** including a plurality of second channels can be formed inside the plurality of second holes **372**, for example, by depositing a metal, a ceramic material, or a silicon material, to fill second holes **372** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Each second channel is a continuous channel that extends vertically inside the stack dies **310**. That is, each second channel has a first end in contact with the bottom surface of the bottommost die and a second end inside the stacked dies without contacting the top surface of the topmost die of the stacked dies. The first end of each second channel of the second contact structure **385** has a third lateral dimension (e.g., x-direction), and the third lateral dimension is smaller than the first lateral dimension of the first end of each first channel. In some implementations, the third lateral dimension is less than 2 μm. In some implementations, the plurality of second channels have different dimensions along the vertical direction. For example, second channel **385-1** has a first end in contact with the bottom surface of die **302** and a second end inside the die **302** without contacting the top surface of die **302**; second channel **385-2** has a first end in contact with the bottom surface of die **302** and a second end inside die **303** without contacting the top surface of die **303**; second channel **385-3** has a first end in contact with the bottom surface of die **302** and a second end inside die **303** without contacting the top surface of die **303**; and second channel **385-4** has a first end in contact with the top surface of die **302** and a second end inside die **305** without contacting the top surface of die **305**. In some implementations, the material of each second channel is one of a metal, a ceramic material, or a silicon material.

[0120] In some implementations, the first contact structure **380** and the second contact structure **385** are formed in a same process; or the first contact structure **380** and the second contact structure **385** can be formed in different processes. The first contact structure **380** and the second contact structure **385** can be made from a same material; or the first contact structure **380** and the second contact structure **385** can be made from different materials. In some implementations, the first channel **382** can include any suitable thermal conductive materials, such as copper (Cu), nickel (Ni), gold (Au), silver (Ag), platinum (Pt), cobalt (Co), titanium (Ti), chrome (Cr), zirconium (Zr), molybdenum (Mo), ruthenium (Ru), hafnium (Hf), tungsten (W), rhenium (Re), graphite, carbon black, combinations thereof, and/or other materials known to those skilled in the art. The second channel **385-1** can comprise any suitable conductive materials, such as Ag, Cu, aluminum (Al), aluminum nitride, silicon carbide, W, graphite, zinc (Zn), combinations thereof, and/or other materials known to those skilled in the art. In some implementations, a ratio between a first width of the first channel **382** and a second width of the second channel **385-1** can be in a range between about 1.5:1 to about 2:1.

[0121] The first contact structure **380** and the second contact structure **385** are then polished such that the surface of the first contact structure **380** and the surface of the second contact structure **385** are flush with the bottom surface of die **302**. In some implementations, a chemical-mechanical polishing (CMP) is performed to polish the first contact structure **380** and the second contact structure **385**.

[0122] Referring back to FIG. **7**, the method **700** can proceed to operation **704**, in which a base die is bonded with the stacked dies. FIG. **3**D illustrates a schematic diagram in a perspective side view of an exemplary semiconductor device after operation **704** of the method **700** shown in FIG. **7**, in accordance with some implementations of the present disclosure.

[0123] As shown in FIG. **3**D, a base die **340** can be bonded with a first surface (bottom surface) of the stacked dies **310** through hybrid bonding or direct bonding, such that a surface of the base die **340** is in contact with the first end of each first channel **382** of the first contact structure **380** and in contact with the first end of each second channel of second contact structure **385**. The base die **340** can be bonded with the stacked dies **310** through hybrid bonding or direct bonding, which is not limited herein. The base die **340** can be provided with wiring structure **344**, which includes first wiring structure **346** and second wiring structure **348** embedded in the base die **340**. The first wiring structure **346** can include any suitable thermal conductive interconnection structures, such as conductive vias and patterned conductive layers, etc., and is configured to transport heat. The first wiring structure **346** is physically connected with the first contact structure **380**. For example, as shown in FIG. **3**D, the first wiring structure **346** is physically connected with the first channel **382**. The second wiring structure **348** is physically connected with the second contact structure **385**. For example, as shown in FIG. **3**D, the second wiring structure **348** is physically connected with the second channel **385-1**. The second wiring structure **348** can include any suitable conductive interconnection structures, such as conductive channels, and configured to transmit electronic signals. In some implementations, the second wiring structure **348** can be configured to transmit both electronic signals and heat. The first wiring structure **346** and the second wiring structure **348** can be isolated from each other.

[0124] In some implementations, the first wiring structure **346** can include any suitable thermal conductive materials, such as copper (Cu), nickel (Ni), gold (Au), silver (Ag), platinum (Pt), cobalt (Co), titanium (Ti), chrome (Cr), zirconium (Zr), molybdenum (Mo), ruthenium (Ru), hafnium (Hf), tungsten (W), rhenium (Re), graphite, carbon black, combinations thereof, and/or other materials known to those skilled in the art. The second wiring structure **348** can comprise any suitable conductive materials, such as Ag, Cu, aluminum (Al), aluminum nitride, silicon carbide, W, graphite, zinc (Zn), combinations thereof, and/or other materials known to those skilled in the art. In some implementations, a ratio between a first wiring width of the first wiring structure **346** and a second wiring width of the second wiring structure **348** can be in a range between about 1.5:1 to about 2:1.

[0125] Referring back to FIG. **7**, the method **700** can proceed to operation **706**, in which an interposer is bonded with the base die. FIG. **3**E illustrates a schematic diagram in a perspective side view of an exemplary semiconductor device after operation **706** of the method **700** shown in FIG. **7**, in accordance with some implementations of the present disclosure.

[0126] As shown in FIG. **3**E, an interposer **320** can be bonded with a surface (bottom surface) of the base die **340**, such that a surface of the interposer **320** is in contact with the wiring structure **344**. The interposer **320** can be bonded with the base die **340** through hybrid bonding or direct bonding, which is not limited herein. The interposer **320** can be any suitable semiconductor material having any suitable structure, such as a monocrystalline single-layer material, a polycrystalline silicon (polysilicon) single-layer material, a polysilicon and metal multi-layer material, etc. The interposer **320** can be provided with connecting structure **324**, which includes first connecting structure **326** and second connecting structure **328** embedded in the interposer **320**. The first connecting structure **326** can include any suitable thermal conductive interconnection structures, such as conductive vias and patterned conductive layers, etc., and is configured to transport heat. The second connecting structure **328** can include any suitable conductive interconnection structures, such as conductive channels, and is configured to transmit electronic signals. In some implementations, the second connecting structure **328** can be configured to transmit both electronic signals and heat. The first connecting structure **326** and the second connecting structure **328** can be isolated from each other.

[0127] Referring back to FIG. **7**, the method **700** can proceed to operation **708**, in which a ball grid array (BGA) is formed on a side of the interposer opposite to the base die. FIG. **3**F illustrates a schematic diagram in a perspective side view of an exemplary semiconductor device after operation **708** of the method **700** shown in FIG. **7**, in accordance with some implementations of the present disclosure.

[0128] As shown in FIG. **7**F, a ball grid array (BGA) **330** is formed on a side of the interposer **320** opposite to the base die **340**. The BGA **330** includes thermal solder balls **332** that are physically connected with the first connecting structure **326** and signal solder balls **335** that are physically connected with the second connecting structure **328**. The thermal solder balls **332** and the signal solder balls **335** can include same materials, and can be formed in a same process. For example, the thermal solder balls **332** and the signal solder balls **335** can include any suitable metal material, such as Aluminum (Al), Antimony (Sb), Arsenic (As), Bismuth (Bi), Cadmium (Cd), Co, Cu, Ni, Au, Ag, Indium (In), Iron (Fe), Lead (Pb), Phosphorus (P), Tin (Sn), Sulfur(S), Zinc (Zn), Germanium (Ge), etc., and any suitable alloy thereof. In some other implementations, the thermal solder balls **332** and the signal solder balls **335** can include different materials. For example, the thermal solder balls **332** can include a first material with a high thermal conductive coefficient, while the signal solder balls **335** can include a second material with a high electric conductive coefficient.

[0129] Referring to FIG. **8**, a flow diagram of an exemplary method for forming a third semiconductor device is illustrated in accordance with some implementations of the present disclosure. It should be understood that the operations and/or steps shown in FIG. **8** are not exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. FIGS. **4**A-**4**G illustrate schematic diagrams in a perspective side view of an exemplary semiconductor device at certain fabricating steps of the method shown in FIG. **8**, in accordance with some implementations of the present disclosure.

[0130] As shown in FIG. **8**, the method **800** starts at operation **802**, in which a plurality of dies are stacked along a vertical direction. A first contact structure is formed in the stacked dies. The first contact structure includes a plurality of first channels, each extending vertically through the stacked dies. FIGS. **4**A-**4**C illustrate schematic diagrams in a perspective side view of exemplary semiconductor devices during operation **802** of the method **800** shown in FIG. **8**, in accordance with some implementations of the present disclosure.

[0131] As shown in FIG. **4**A, in some implementations, a plurality of dies are stacked through direct bonding, which forms bonding between surfaces without using intermediate layers, such as solder or adhesives, in a vertical direction on a carrier wafer **415**. Exemplarily, as shown in FIG. **4**A, die **402**, die **403**, die **404**, and die **405** are stacked vertically on the carrier wafer **415** to form stacked dies **410**. In some implementations, the stacked dies **410** can be any suitable semiconductor dies/die including one or more memory dies. The stacked dies **410** can include low power chips having maximum operation powers lower than a threshold power value. The stacked dies **410** can include high power chips having maximum operation powers higher than the threshold power value. For example, the stacked dies **410** include at least one of a micro processing chip, a logic control chip, a power management chip, a driver chip, and an analog chip, and may also include at least one of a memory chip and a sensing chip.

[0132] As shown in FIG. **4**B, in some implementations, a sacrificial layer **412** is deposited on a first surface of the stacked dies **410**, and a plurality of first holes **470** are formed in the sacrificial layer **412** and vertically extend through the stacked dies **410**. In some implementations, the carrier wafer **415** is removed before forming the sacrificial layer **412**. In some implementations, the carrier wafer **415** is used as the sacrificial layer **412** and the plurality of first holes **470** are formed in the carrier wafer **415**. A lithography process can be performed to pattern the first holes using an etch mask (e.g., a photoresist mask and/or a hard mask), and one or more dry etching and/or wet etching processes, such as RIE, can be performed to etch first holes **470** in the sacrificial layer **412**. Thus, first holes **470** extending vertically in the sacrificial layer **412** can be formed.

[0133] As shown in FIG. **4**B, in some implementations, a plurality of second holes **472** are formed in the sacrificial layer **412** and vertically extend inside the stacked dies **410**. A lithography process can be performed to pattern the second holes **472** using an etch mask (e.g., a photoresist mask and/or a hard mask), and one or more dry etching and/or wet etching processes, such as RIE, can be performed to etch second holes **472** in the sacrificial layer **412**. Thus, second holes **472** extending vertically in the sacrificial layer **412** can be formed. In some implementations, the plurality of second holes **472** have different dimensions along the vertical direction. The dimensions of the second holes **472** along the vertical direction can be controlled by etching each second hole **472** with different time durations or by back-filling the second holes **472** to achieve a certain vertical dimension. In some implementations, forming the second holes **472** can be performed in a same process of forming the first holes **470**. A lateral dimension (e.g., the x-direction) of each first hole **470** is larger than a lateral dimension (e.g., the x-direction) of each second hole **472**. In some implementations, the second holes **472** can be sandwiched between two sets of first holes **470**, or the second holes **472** can be surrounded by the plurality of first holes **470**.

[0134] As shown in FIG. **4**C, in some embodiments, a first contact structure **480** including a plurality of first channels **482** is formed inside the plurality of first holes **470**, for example, by depositing a metal, a ceramic material, or a silicon material, to fill first holes **470** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Each first channel **482** is a continuous channel that extends through the stacked dies **410**. That is, each first channel **482** has a first end in contact with the bottom surface of the bottommost die and a second end in contact with the top surface of the topmost die of the stacked dies **410**. The first end of each first channel **482** has a first lateral dimension (e.g., x-direction), the second end of each first channel **482** has a second lateral dimension (e.g., x-direction), and the first lateral dimension is larger than the second lateral dimension. In some implementations, the first lateral dimension is about 2-3 μm. The material of each first channel **482** has a thermal conductive coefficient of no less than 20 W/mK. In some implementations, the material of each first channel **482** is one of a metal, a ceramic material, or a silicon material. In some implementations, the material of each first channel **482** can be one of silicon nitride, aluminum oxide, and silicon carbide. A planarization process, such as CMP, can be performed to remove excess first channel **482** that is deposited beyond the top surface of the topmost die.

[0135] In some implementations, as shown in FIG. **4**C, a second contact structure **485** including a plurality of second channels can be formed inside the plurality of second holes **472**, for example, by depositing a metal, a ceramic material, or a silicon material, to fill second holes **472** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Each second channel is a continuous channel that extends vertically inside the stack dies **410**. That is, each second channel has a first end in contact with the bottom surface of the bottommost die and a second end inside the stacked dies without contacting the top surface of the topmost die of the stacked dies. The first end of each second channel of the second contact structure **485** has a third lateral dimension (e.g., x-direction), and the third lateral dimension is smaller than the first lateral dimension of the first end of each first channel. In some implementations, the third lateral dimension is less than 2 μm. In some implementations, the plurality of second channels have different dimensions along the vertical direction. For example, second channel **485-1** has a first end in contact with the bottom surface of die **402** and a second end inside the die **402** without contacting the top surface of die **402**; second channel **485-2** has a first end in contact with the bottom surface of die **402** and a second end inside die **403** without contacting the top surface of die **403**; second channel **485-3** has a first end in contact with the bottom surface of die **402** and a second end inside die **403** without contacting the top surface of die **403**; and second channel **485-4** has a first end in contact with the top surface of die **402** and a second end inside die **405** without contacting the top surface of die **405**. In some implementations, a material of each second channel is one of a metal, a ceramic material, or a silicon material.

[0136] In some implementations, the first contact structure **480** and the second contact structure **485** are formed in a same process; or the first contact structure **480** and the second contact structure **485** can be formed in different processes. The first contact structure **480** and the second contact structure **485** can be made from a same material; or the first contact structure **480** and the second contact structure **485** can be made from different materials. In some implementations, the first channel **482** can include any suitable thermal conductive materials, such as copper (Cu), nickel (Ni), gold (Au), silver (Ag), platinum (Pt), cobalt (Co), titanium (Ti), chrome (Cr), zirconium (Zr), molybdenum (Mo), ruthenium (Ru), hafnium (Hf), tungsten (W), rhenium (Re), graphite, carbon black, combinations thereof, and/or other materials known to those skilled in the art. The second channel **485-1** can comprise any suitable conductive materials, such as Ag, Cu, aluminum (Al), aluminum nitride, silicon carbide, W, graphite, zinc (Zn), combinations thereof, and/or other materials known to those skilled in the art. In some implementations, a ratio between a first width of the first channel **482** and a second width of the second channel **485-1** can be in a range between about 1.5:1 to about 2:1.

[0137] The first contact structure **480** and the second contact structure **485** are then polished such that the surface of the first contact structure **480** and the surface of the second contact structure **485** are flush with the bottom surface of die **402**. In some implementations, a chemical-mechanical polishing (CMP) is performed to polish the first contact structure **480** and the second contact structure **485**.

[0138] Referring back to FIG. **8**, the method **800** can proceed to operation **804**, in which a base die is bonded with the stacked dies. FIG. **4**D illustrates a schematic diagram in a perspective side view of an exemplary semiconductor device after operation **804** of the method **800** shown in FIG. **8**, in accordance with some implementations of the present disclosure.

[0139] As shown in FIG. **4**D, a base die **440** can be bonded with a first surface (bottom surface) of the stacked dies **410** through hybrid bonding or direct bonding, such that a surface of the base die **440** is in contact with the first end of each first channel **482** of the first contact structure **480** and in contact with the first end of each second channel of second contact structure **485**. The base die **440** can be bonded with the stacked dies **410** through hybrid bonding or direct bonding, which is not limited herein. The base die **440** can be provided with wiring structure **444**, which includes first wiring structure **446** and second wiring structure **448** embedded in the base die **440**. The first wiring structure **446** can include any suitable thermal conductive interconnection structures, such as conductive vias and patterned conductive layers, etc., and is configured to transport heat. The first wiring structure **446** is physically connected with the first contact structure **480**. For example, as shown in FIG. **4**D, the first wiring structure **446** is physically connected with the first channel **482**. The second wiring structure **448** is physically connected with the second contact structure **485**. For example, as shown in FIG. **4**D, the second wiring structure **448** is physically connected with the second channel **485-1**. The second wiring structure **448** can include any suitable conductive interconnection structures, such as conductive channels, and configured to transmit electronic signals. In some implementations, the second wiring structure **448** can be configured to transmit both electronic signals and heat. The first wiring structure **446** and the second wiring structure **448** can be isolated from each other.

[0140] In some implementations, the first wiring structure **446** can include any suitable thermal conductive materials, such as copper (Cu), nickel (Ni), gold (Au), silver (Ag), platinum (Pt), cobalt (Co), titanium (Ti), chrome (Cr), zirconium (Zr), molybdenum (Mo), ruthenium (Ru), hafnium (Hf), tungsten (W), rhenium (Re), graphite, carbon black, combinations thereof, and/or other materials known to those skilled in the art. The second wiring structure **448**

can comprise any suitable conductive materials, such as Ag, Cu, aluminum (Al), aluminum nitride, silicon carbide, W, graphite, zinc (Zn), combinations thereof, and/or other materials known to those skilled in the art. In some implementations, a ratio between a first wiring width of the first wiring structure **446** and a second wiring width of the second wiring structure **448** can be in a range between about 1.5:1 to about 2:1.

[0141] Referring back to FIG. **8**, the method **800** can proceed to operation **806**, in which an interposer is bonded with the base die. FIG. **4**E illustrates a schematic diagram in a perspective side view of an exemplary semiconductor device after operation **806** of the method **800** shown in FIG. **8**, in accordance with some implementations of the present disclosure.

[0142] As shown in FIG. **4**E, an interposer **420** can be bonded with a surface (bottom surface) of the base die **440**, such that a surface of the interposer **420** is in contact with the wiring structure **444**. The interposer **420** can be bonded with the base die **440** through hybrid bonding or direct bonding, which is not limited herein. The interposer **420** can be any suitable semiconductor material having any suitable structure, such as a monocrystalline single-layer material, a polycrystalline silicon (polysilicon) single-layer material, a polysilicon and metal multi-layer material, etc. The interposer **420** can be provided with connecting structure **424**, which includes first connecting structure **426** and second connecting structure **428** embedded in the interposer **420**. The first connecting structure **426** can include any suitable thermal conductive interconnection structures, such as conductive vias and patterned conductive layers, etc., and is configured to transport heat. The second connecting structure **428** can include any suitable conductive interconnection structures, such as conductive channels, and is configured to transmit electronic signals. In some implementations, the second connecting structure **428** can be configured to transmit both electronic signals and heat. The first connecting structure **426** and the second connecting structure **328** can be isolated from each other.

[0143] Referring back to FIG. **8**, the method **800** can proceed to operation **808**, in which a ball grid array (BGA) is formed on a side of the interposer opposite to the base die. FIG. **4**F illustrates a schematic diagram in a perspective side view of an exemplary semiconductor device after operation **808** of the method **800** shown in FIG. **8**, in accordance with some implementations of the present disclosure.

[0144] As shown in FIG. **4**F, a ball grid array (BGA) **430** is formed on a side of the interposer **420** opposite to the base die **440**. The BGA **430** includes thermal solder balls **432** that are physically connected with the first connecting structure **426** and signal solder balls **435** that are physically connected with the second connecting structure **428**. The thermal solder balls **432** and the signal solder balls **435** can include same materials, and can be formed in a same process. For example, the thermal solder balls **432** and the signal solder balls **435** can include any suitable metal material, such as Aluminum (Al), Antimony (Sb), Arsenic (As), Bismuth (Bi), Cadmium (Cd), Co, Cu, Ni, Au, Ag, Indium (In), Iron (Fe), Lead (Pb), Phosphorus (P), Tin (Sn), Sulfur(S), Zinc (Zn), Germanium (Ge), etc., and any suitable alloy thereof. In some other implementations, the thermal solder balls **432** and the signal solder balls **435** can include different materials. For example, the thermal solder balls **432** can include a first material with a high thermal conductive coefficient, while the signal solder balls **435** can include a second material with a high electric conductive coefficient.

[0145] Referring back to FIG. **8**, the method **800** can proceed to operation **810**, in which a thermal conductive layer is formed on a second surface (top surface) of the stacked dies. The second surface of the stacked dies is opposite to the first surface of the stacked dies in the vertical direction. FIG. **4**G illustrates a schematic diagram in a perspective side view of an exemplary semiconductor device after operation **810** of the method **800** shown in FIG. **8**, in accordance with some implementations of the present disclosure.

[0146] As shown in FIG. **4**G, a thermal conductive layer **406** can be formed by depositing a thermal conductive material to cover the top surface of the topmost die of the stacked dies **410**, so that the thermal conductive layer **406** is in contact with each first channel **482**. A material of the thermal conductive layer **406** has a thermal conductive coefficient of no less than 20 W/mK. In some implementations, the material of the thermal conductive layer **406** is one of a metal, a ceramic material, or a silicon material. In some implementations, the material of each first channel can be one of silicon nitride, aluminum oxide, and silicon carbide. In some implementations, the thermal conductive layer **406** can include any suitable thermal conductive materials, such as copper (Cu), nickel (Ni), gold (Au), silver (Ag), platinum (Pt), cobalt (Co), titanium (Ti), chrome (Cr), zirconium (Zr), molybdenum (Mo), ruthenium (Ru), hafnium (Hf), tungsten (W), rhenium (Re), graphite, carbon black, combinations thereof, and/or other materials known to those skilled in the art.

[0147] This disclosure further provides a system, as described in International Application No. PCT/CN2024/078561, filed on Feb. 26, 2024, which is incorporated by reference herein. FIG. **9** illustrates a block diagram of an example system **900** having one or more semiconductor devices (e.g., memory devices), in accordance with some implementations of the present disclosure. The system **900** can be a mobile phone, a desktop computer, a laptop computer, a tablet, a vehicle computer, a gaming console, a printer, a positioning device, a wearable electronic device, a smart sensor, a virtual reality (VR) device, an argument reality (AR) device, or any other suitable electronic devices having storage therein. As shown in FIG. **9**, the system **900** can include one or more memory die **902**, a base device **904**, a computing device **908**, and an external host device **912**. In some implementations, each of devices **902**, **904**, **908**, and **912** can be a die or multiple dies stacked together. Each of devices **902**, **904**, **908**, and **912** can be manufactured by depositing multiple layers of various materials and etching them onto a semiconductor wafer in intricate patterns defined by a chip design. After the wafer fabrication process is complete, the wafer that includes individual circuits is cut and diced into individual pieces, each of which is a die. Each die can include a fully functional electronic circuit, which can be a microprocessor, memory, sensor, or any other suitable type of integrated circuit. In some embodiments, each die is encapsulated in a protective package, providing physical support, protection from environmental factors, and connections (e.g., through pins or solder balls) to external devices or system.

[0148] Memory die **902** can include any memory device disclosed herein, such as a memory device (e.g., a 3D memory device) based on any one of semiconductor structures as described with respect to FIGS. **1A-1C**, FIGS. **2A-2F**, FIGS. **3A-3F**, and FIGS. **4A-4G**. In some implementations, memory die **902** includes one or more dynamic random-access memory (DRAM) devices. In some implementations, memory die **902** includes one or more NAND Flash memories. In some implementations, memory die **902** can include a high bandwidth memory (HBM). In some implementations, memory die **902** can be stacked together, e.g., as described with further details with respect to FIGS. **1A-1C**, FIGS. **2A-2F**, FIGS. **3A-3F**, and FIGS. **4A-4G**. In some implementations, memory die **902** can include a combination of one or more HBM devices.

[0149] Base die **904** (also referred to as a logic die or a buffer die) can include buffer circuitry and test logic for memory die **902**. Base die **904** can be configured to provide physical layer communication protocols (e.g., IEEE-1500) between memory die **902** and computing die **908**. Base die **904** can be configured to transmit data between memory die **902** and computing die **908** based on control commands and addresses from computing die **908**.

[0150] Computing die **908** can be a logic device and can include at least one processor of an electronic device, such as a central processing unit (CPU), a graphics processing unit (GPU), an application-specific integrated circuit (ASIC), or a system-on-chip (SoC), such as an application processor (AP). Computing die **908** can be configured to send or receive data to or from memory die **902**. Computing die **908** is coupled to base die **904** through an interface (IF) **906**. Interface **906** can include connections provided by bonding contacts or an interposer (e.g., as described with respect to FIGS. **1A-1C**, FIGS. **2A-2F**, FIGS. **3A-3F**, and FIGS. **4A-4G**). In some implementations, interface **906** includes connections provided by any suitable combination of the aforementioned techniques.

[0151] System **900** can further include the external host die **912** coupled to computing die **908** through an interface (IF) **910**. For example, external host die **912** can be a computer, and computing die **908** can be a CPU of the computer. In this example, interface **910** includes connections provided by a mainboard of the computer that are coupled to the CPU. As another example, external host die **912** is a graphics card, computing die **908** is a GPU of the graphics card, and interface **910** includes connections provided by a printed circuit board (PCB) of the graphics card that are coupled to the GPU.

[0152] System **900** may further include a memory controller (a.k.a., a controller circuit, which is not shown in FIG. **9**) coupled to memory die **902**. In some implementations, the memory controller is located in the computing die **908**. Consistent with implementations of the present disclosure, the memory controller can include conductive interconnections through a cover layer that are in contact with conductive pads in a conductive pad layer, and the memory controller can be coupled to memory die **902** through at least one of the conductive interconnections. The memory controller is configured to control memory die **902**. For example, the memory controller may be configured to operate channel structures via word lines. The memory controller can manage data stored in memory die **902** and communicate with computing die **908**.

[0153] In some implementations, the memory controller is designed/configured for operating in a low duty-cycle environment like secure digital (SD) cards, compact Flash (CF) cards, universal serial bus (USB) Flash drives, or other media for use in electronic devices, such as personal computers, digital cameras, mobile phones, etc. In some implementations, the memory controller is designed/configured for operating in a high duty cycle environment SSDs or embedded multi-media-cards (eMMCs) used as data storage for mobile devices, such as smartphones, tablets, laptop computers, etc., and enterprise storage arrays. The memory controller can be configured to control operations of memory die **902**, such as read, erase, and program (or write) operations. The memory controller can also be configured to manage various functions with respect to the data stored or to be stored in memory die **902** including, but not limited to bad-block management, garbage collection, logical-to-physical address conversion, wear leveling, etc. In some implementations, the memory controller is further configured to process error correction codes (ECCs) with respect to the data read from or written to memory die **902**. In some other implementations, the base die **904** instead of the memory controller is configured to process ECCs. Any other suitable functions may be performed by the memory controller as well, for example, formatting memory die **902**.

[0154] The memory controller can communicate with an external device (e.g., computing die **908**) according to a particular communication protocol. For example, the memory controller may communicate with the external device through at least one of various interface protocols, such as a USB protocol, an MMC protocol, a peripheral component interconnection (PCI) protocol, a PCIexpress (PCTe or PCI-e) protocol, an advanced technology attachment (ATA) protocol, a serial-ATA protocol, a parallel-ATA protocol, a small computer small interface (SCSI) protocol, an enhanced small disk interface (ESDI) protocol, an integrated drive electronics (IDE) protocol, a Firewire protocol, etc.

[0155] The memory controller and one or more memory die **902** can be integrated into various types of storage devices, for example, be included in the same package, such as a universal Flash storage (UFS) package or an eMMC package. That is, system **900** can be implemented and packaged into different types of end electronic products. For example, the memory controller and a single memory die **902** may be integrated into a memory card. The memory card can include a PC card (PCMCIA, personal computer memory card international association), a CF card, a smart media (SM) card, a memory stick, a multimedia card (MMC, RS-MMC, MMCmicro), an SD card (SD, miniSD, microSD, SDHC), a UFS, etc.

[0156] It is noted that, although not shown, the above descripted methods can further comprise any other suitable operations to further form semiconductor devices. For example, the formed semiconductor devices can be attached to a printed circuit board (PCB).

[0157] The present disclosure provides semiconductor devices with a high thermal conductivity. In the era of great computing power and artificial intelligence (AI), the von Neumann architecture that combines high-bandwidth memory (HBM) with GPU remains the mainstream in the high-performance chip market. However, the high amount of heat generated by the stacked HBM is an urgent issue that needs to be addressed. In the traditional HBM architecture, as the number of stacked DRAM layers increases, the heat dissipation for different layers of DRAM varies, which can easily lead to slow heat dissipation for DRAMs located in the middle, causing a decline in performance. Because

DRAM needs to continuously perform read and write operations, it tends to generate heat during operation. The stacked structure of HBM limits the heat dissipation of the DRAM located in the middle layers.

[0158] In the semiconductor devices provided by this disclosure, the heat generated during the operation of HBM can be rapidly dissipated through first channels to prevent localized overheating. The number of I/O channels can be significantly increased. The first channels have relatively large lateral dimensions, extend through the entire stacked dies, do not carry out data transmission functions, and act as a thermal conduit to transfer heat. Compared to prior art, the current solution achieves better internal heat dissipation inside stacked dies without increasing the chip area of the HBM.

[0159] The foregoing description of the specific implementations will so fully reveal the general nature of the present disclosure that others can, by applying knowledge within the skill of the art, readily modify and/or adapt, for various applications, such specific implementations, without undue experimentation, and without departing from the general concept of the present disclosure. Therefore, such adaptations and modifications are intended to be within the meaning and range of equivalents of the disclosed implementations, based on the disclosure and guidance presented herein. It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by the skilled artisan in light of the disclosure and guidance.

[0160] Implementations of the present disclosure have been described above with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries can be defined so long as the specified functions and relationships thereof are appropriately performed.

[0161] The Summary and Abstract sections can set forth one or more but not all exemplary implementations of the present disclosure as contemplated by the inventor(s), and thus, are not intended to limit the present disclosure and the appended claims in any way.

[0162] The breadth and scope of the present disclosure should not be limited by any of the above-described exemplary implementations, but should be defined only in accordance with the following claims and their equivalents.

What is claimed is:

1. A semiconductor device, comprising:

a plurality of dies stacked in a vertical direction;

a thermal conductive layer deposited on a top surface of a topmost die;

a base die located on a bottom surface of a bottommost die; and

a first contact structure extending vertically through the plurality of dies,

wherein the first contact structure comprises one or more first channels, each first channel having a first end being in contact with the base die and having a second end being in contact with the thermal conductive layer.

2. The semiconductor device according to claim **1**, further comprising a second contact structure extending vertically inside the plurality of dies, wherein the second contact structure comprises one or more second channels, each second channel extending vertically and being in contact with the base die without contacting the thermal conductive layer.

3. The semiconductor device according to claim **2**, wherein the base die comprises a wiring structure including a first wiring structure and a second wiring structure, wherein the first wiring structure is configured to transport heat and the second wiring structure is configured to transmit electronic signals.

4. The semiconductor device according to claim **3**, further comprising:

an interposer on a side of the base die opposite to the plurality of dies, wherein the interposer comprises a ball grid array on a side of the interposer opposite to the base dies.

5. The semiconductor device according to claim **4**, wherein the ball grid array comprises:

one or more thermal solder balls in connection with the first wiring structure; and

one or more signal solder balls in connection with the second wiring structure.

6. The semiconductor device according to claim **2**, wherein each of the first channel and the second channel is a continuous structure along the vertical direction.

7. The semiconductor device according to claim **2**, wherein the second contact structure is surrounded by the first contact structure.

8. A semiconductor device, comprising:

a plurality of dies stacked in a vertical direction;

an interposer located on a bottom surface of a base die, wherein the interposer comprises a wiring structure including a first wiring structure and a second wiring structure; and

a first contact structure extending vertically through the plurality of dies, wherein the first contact structure comprises one or more first channels, each first channel having a first end being in contact with the interposer and having a second end being in contact with a top surface of a topmost die.

9. The semiconductor device according to claim **8**, further comprising:

a second contact structure extending vertically inside the plurality of dies, wherein the second contact structure comprises one or more second channels, each second channel extending vertically and being in contact with the interposer without contacting the top surface of the topmost die.

10. The semiconductor device according to claim **8**, further comprising:

a ball grid array on a side of the interposer opposite to the plurality of dies.

11. The semiconductor device according to claim **10**, wherein the ball grid array comprises:

one or more thermal solder balls in contact with the first wiring structure, wherein the first wiring structure is configured to transport heat; and

one or more signal solder balls in contact with the second wiring structure, wherein the second wiring structure is configured to transmit electronic signals.

12. The semiconductor device according to claim **9**, wherein each of the first channel and the second channel is a continuous structure along the vertical direction.

13. The semiconductor device according to claim **9**, wherein the second contact structure is surrounded by the first contact structure.

14. A method of forming a semiconductor device, comprising:

stacking a plurality of dies in a vertical direction;

depositing a sacrificial layer on a first surface of the stacked dies, and forming a plurality of first holes in the sacrificial layer;

forming, using the plurality of first holes, a first contact structure extending vertically through the stacked dies, wherein the first contact structure comprises one or more first channels;

bonding a base die on the first surface of the stacked dies through hybrid bonding;

bonding an interposer on the base die, wherein the interposer comprises a wiring structure;

connecting each first channel with the wiring structure; and

forming a thermal conductive layer on a second surface of the stacked die, wherein the second surface is opposite to the first surface in the vertical direction, and the thermal conductive layer is physically connected with each first channel.

15. The method according to claim **14**, further comprising:

forming a plurality of second holes in the sacrificial layer;

forming, using the plurality of second holes, a second contact structure extending vertically inside the stacked dies, wherein the second contact structure comprises one or more second channels; and

connecting each second channel with the interposer without physically contacting the thermal conductive layer.

16. The method according to claim **15**, further comprising:

forming a ball grid array on a side of the interposer opposite to the stacked dies.

17. The method according to claim **16**, wherein the wiring structure comprises a first wiring structure configured to transport heat and a second wiring structure configured to transmit electronic signals, and forming the ball grid array comprises:

forming a plurality of thermal solder balls in contract with the first wiring structure; and

forming a plurality of signal solder balls in contact with the second wiring structure.

18. The method according to claim **15**, wherein forming the first contact structure comprises forming each first channel continuously extending along the vertical direction; and

forming the second contact structure comprises forming each second channel continuously extending along the vertical direction.

19. The method according to claim **15**, wherein forming the second contact structure comprises forming the second contact structure surrounded by the first contact structure.

20. The method according to claim **15**, further comprising forming each first channel with a first end having a first lateral dimension and a second end having a second lateral dimension, wherein the first end of each first channel is in contact with the base die, the second end of each first channel is in contact with the thermal conductive layer, and the first lateral dimension is larger than the second lateral dimension.

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